



Supplementary Materials for

Direct observation of chirality-induced spin selectivity in electron donor–acceptor molecules

Hannah J. Eckvahl *et al.*

Corresponding authors: Stefano Carretta, stefano.carretta@unipr.it; Matthew D. Krzyaniak, mdkrzyaniak@northwestern.edu;
Michael R. Wasielewski, m-wasielewski@northwestern.edu

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Materials and Methods

Synthesis

The synthetic schemes and structures of the intermediates and final structures are illustrated in Figs. S1-S2. All commercially available materials were used without further purification unless
5 otherwise noted. Characterization of synthesized compounds was performed at Integrated Molecular Structure Education and Research Center (IMSERC) at Northwestern University. Proton and carbon NMR spectra were recorded on a Bruker Avance III 500 MHz system equipped with DCH CryoProbe. Shifts are reported in parts-per-million (ppm) relative to TMS. High Resolution Mass Spectra (HRMS) were obtained with an Agilent LCTOF 6200 series mass
10 spectrometer using electrospray ionization (ESI), APCI, and APPI. CD spectra were obtained with Jasco J-815 CD Instrument Keck Biophysics Facility at Northwestern University.

NIA-xyl. In a 250 mL two-neck round bottom flask equipped with a condenser and a dropping funnel, 1,4,5,8-naphthalenetetracarboxylic dianhydride (3.0 g, 11.2 mmol, regenerated prior to use by refluxing overnight in acetic anhydride followed by the removal of the acetic anhydride under
15 reduced pressure) was suspended in 40 mL of dry *N,N*-dimethylformamide. The mixture was heated to reflux under an inert atmosphere and stirred for two hours to ensure the dissolution of the 1,4,5,8-naphthalenetetracarboxylic dianhydride. To this vigorously stirred solution, 2,6-xylidine (677 mg, 5.6 mmol) in 50 ml of dry *N,N*-dimethylformamide was added via a dropping funnel over several hours. The reaction mixture was refluxed overnight under an inert atmosphere,
20 cooled down to room temperature, and the solvent removed under reduced pressure. The crude material was purified by silica gel chromatography (100 % dichloromethane), affording **NIA-xyl** as a brown powder (800 mg, 2.15 mmol, 38 %). ¹H NMR (500 MHz, CD₂Cl₂, δ): 8.76-8.80 (m, 4H), 7.27 (m, 1H), 7.19 (m, 2H), 2.05 (m, 6H). ¹³C NMR (126 MHz, CDCl₃, δ): 161.43, 160.90, 158.25, 135.00, 134.92, 132.76, 132.47, 132.38, 130.81, 130.60, 128.42, 128.30, 128.26, 127.85,

127.80, 127.04, 126.85, 126.76, 126.14, 122.36, 16.80. ESI-HRMS (m/z): calculated for $C_{23}H_9N_5O$ $[M]^+$: 371.0813, found: 371.0803.

NH₂-NDI-xyI. In a 25 mL round bottom flask equipped with a condenser, **NIA-xyI** (300 mg, 0.81 mmol) was suspended in 10 mL of dry *N,N*-dimethylformamide. The mixture was heated to reflux and stirred for one hour. Hydrazine monohydrate (0.5 mL) was added, and the reaction mixture was refluxed for 3 hours. The reaction mixture was cooled to room temperature and the solvent was removed under reduced pressure. The crude material was dissolved in dichloromethane and washed twice with brine. The organic layer was dried over sodium sulfate, filtered, and concentrated under reduced pressure. The crude material was purified by silica gel chromatography (20 % ethyl acetate in dichloromethane), affording **NH₂-NDI-xyI** as a yellow solid (270 mg, 0.70 mmol, 87 %). ¹H NMR (500 MHz, CD₂Cl₂, δ): 8.71-8.76 (m, 4H), 7.26 (m, 1H), 7.18 (d, J = 7.7 Hz, 2H), 5.52 (s, 2H), 2.04 (s, 6H). ¹³C NMR (126 MHz, CD₂Cl₂, δ): 161.32, 158.81, 135.00, 132.71, 130.58, 130.34, 128.28, 127.78, 126.64, 126.13, 125.49, 124.83, 16.82. ESI-HRMS (m/z): calculated for $C_{22}H_{15}N_3O_4$ $[M+H]^+$: 386.1135, found: 386.1152.

NMA-NMI-NDI-xyI. In a 25 mL two-neck round bottom flask equipped with a condenser and a dropping funnel, 5,5'-dimethoxy-1H,1'H,3H,3'H-[6,6'-bibenzo[de]isochromene]-1,1',3,3'-tetraone(50) (327 mg, 0.72 mmol) was suspended in 10 mL of dry *N,N*-dimethylformamide and the mixture was refluxed for one hour under inert atmosphere. To this, **NH₂-NDI-xyI** (250 mg, 0.65 mmol) dissolved in 50 mL of dry *N,N*-dimethylformamide was added via a dropping funnel over several hours, and the reaction mixture was further refluxed overnight under an inert atmosphere. The reaction mixture was cooled to room temperature and the solvent was removed under reduced pressure. Due to purification difficulties and the tendency of the product to undergo hydrolysis of the anhydride moiety, the crude material was passed through a silica gel pad (10 % ethyl acetate in dichloromethane) and used without further purification right away. A small

fraction of the crude was purified using preparatory TLC for characterization. ^1H NMR (500 MHz, CD_2Cl_2 , δ): 8.84 (m, 4H), 8.55 (s, 1H), 8.47-8.51 (m, 2H), 8.43 (d, $J = 6.8$ Hz, 1H), 7.51-7.57 (m, 2H), 7.28 (m, 1H), 7.21 (d, $J = 7.6$ Hz, 2H), 3.87 (s, 3H), 3.84 (s, 3H), 2.08 (s, 6H). ^{13}C NMR (126 MHz, CD_2Cl_2 , δ): 162.00, 160.53, 160.44, 159.92, 159.86, 156.06, 155.87, 133.44, 132.61, 132.48, 132.22, 132.08, 132.05, 131.46, 131.43, 131.32, 130.47, 129.11, 128.58, 128.29, 128.00, 127.81, 127.56, 127.48, 126.41, 126.10, 125.93, 125.46, 124.25, 123.76, 122.09, 120.58, 119.01, 118.72, 117.95, 29.68, 17.59. APCI-HRMS (m/z): calculated for $\text{C}_{48}\text{H}_{27}\text{N}_3\text{O}_{11}$ $[\text{M}]^+$: 821.1651, found: 821.1627.

Br-Ph-(NMI) $_2$ -NDI-xyl. In a 50 mL round bottom flask equipped with a condenser, **NMA-NMI-NDI-xyl** (210 mg, crude), 4-bromoaniline (132 mg, 0.77 mmol), and zinc (II) acetate dihydrate (15 mg) were dissolved in 20 mL of dry *N,N*-dimethylformamide and the mixture was refluxed overnight under an inert atmosphere. The reaction mixture was cooled to room temperature and the solvent was removed under reduced pressure. The crude material was purified by silica gel chromatography (gradient from 100 % dichloromethane to 5 % ethyl acetate in dichloromethane), affording **Br-Ph-(NMI) $_2$ -NDI-xyl** as a brown solid (40 mg, 0.30 mmol). ^1H NMR (500 MHz, CD_2Cl_2 , δ): 8.80-8.90 (m, 4H), 8.56 (s, 1H), 8.50 (m, 2H), 8.43 (dd, $J_1 = 7.0$ Hz, $J_2 = 1.1$ Hz, 1H), 7.66 (d, $J = 8.6$ Hz, 2H), 7.49-7.58 (m, 3H), 7.45 (dd, $J_1 = 8.6$ Hz, $J_2 = 1.1$ Hz, 1H), 7.29 (m, 1H), 7.18-7.23 (m, 4H), 3.86 (m, 6H), 2.08 (s, 6H). ^{13}C NMR (126 MHz, CD_2Cl_2 , δ): 163.20, 163.01, 161.25, 159.99, 159.96, 159.18, 159.11, 155.13, 155.07, 135.02, 135.01, 134.05, 132.66, 131.99, 131.72, 131.64, 131.57, 131.33, 131.30, 130.72, 130.70, 130.49, 129.87, 129.70, 128.69, 128.36, 127.83, 127.19, 127.13, 127.03, 126.75, 126.70, 125.64, 125.44, 123.90, 123.80, 123.48, 123.34, 122.74, 122.08, 121.75, 121.24, 117.19, 116.37, 28.93, 16.85. ESI-HRMS (m/z): calculated for $\text{C}_{54}\text{H}_{31}\text{BrN}_4\text{O}_{10}\text{Na}$ $[\text{M}+\text{Na}]^+$: 999.1109, found: 999.1125.

(*R,S*)-**1-d9**. In a 25 mL round bottom two-neck flask equipped with a condenser and a septum were combined **Br-Ph-(NMI)₂-NDI-xy1** (20 mg, 0.02 mmol), PXX-*d*₉-Bpin (Fig. S2) (17 mg, 0.04 mmol), [1,1'-bis(diphenylphosphino)ferrocene]dichloropalladium(II) (3 mg, 0.004 mmol), and potassium carbonate (20 mg). The flask was evacuated and filled with nitrogen three times. To this mixture, 10 mL of a degassed mixture of toluene, ethanol, and water (2/1/1 vol. ratio) was added through the septum. The reaction mixture was heated at 110 °C overnight under an inert atmosphere. The reaction mixture was cooled to room temperature, dissolved in dichloromethane, and washed twice with brine. The organic layer was dried over sodium sulfate, filtered, and concentrated under reduced pressure. The crude material was purified by silica gel chromatography (gradient from 100 % dichloromethane to 5 % ethyl acetate in dichloromethane), affording (*R,S*)-**1-d9** as a yellow solid (17 mg, 0.014 mmol, 72 %). ¹H NMR (500 MHz, CD₂Cl₂, δ): 8.81-8.88 (m, 4H), 8.58 (s, 1H), 8.55 (s, 1H), 8.51 (dd, *J*₁ = 6.5 Hz, *J*₂ = 1.5 Hz, 1H), 8.47 (dd, *J*₁ = 7.0 Hz, *J*₂ = 1.0 Hz, 1H), 7.62 (d, *J* = 8.4 Hz, 2H), 7.52-7.60 (m, 3H), 7.47 (dd, *J*₁ = 8.5 Hz, *J*₂ = 1.0 Hz, 1H), 7.41 (d, *J* = 8.2 Hz, 2H), 7.29 (m, 1H), 7.21 (d, *J* = 7.7 Hz, 2H), 3.89 (s, 3H), 3.87 (s, 3H), 2.09 (s, 6H). ¹³C NMR (126 MHz, CD₂Cl₂, δ): 163.48, 163.29, 161.26, 160.02, 159.98, 159.19, 159.12, 155.16, 155.09, 151.74, 151.60, 143.78, 143.36, 139.24, 135.03, 134.10, 132.66, 132.03, 131.65, 131.34, 131.31, 130.72, 130.47, 130.38, 129.72, 129.41, 128.65, 128.35, 128.25, 127.83, 127.20, 127.14, 127.03, 126.76, 125.65, 125.55, 124.03, 123.77, 123.50, 123.40, 122.73, 122.30, 121.25, 121.09, 120.70, 117.22, 116.36, 110.96, 110.56, 28.93, 16.86. APPI-HRMS (*m/z*): calculated for C₇₄H₃₁D₉N₄O₁₂ [M]⁺: 1185.3208, found: 1185.3225.

(*R,S*)-**1-h9**. Using a procedure described for the synthesis of (*R,S*)-**1-d9**, (*R,S*)-**1-h9** was obtained from 20 mg (0.02 mmol) of **D** and 18 mg of PXX-Bpin(3*I*) as a yellow solid (16 mg, 66 %). ¹H NMR (500 MHz, CD₂Cl₂, δ): 8.81-8.88 (m, 4H), 8.58 (s, 1H), 8.55 (s, 1H), 8.51 (dd, *J*₁ = 6.6 Hz, *J*₂ = 1.4 Hz, 1H), 8.47 (dd, *J*₁ = 7.0 Hz, *J*₂ = 0.7 Hz, 1H), 7.63 (d, *J* = 8.2 Hz, 2H), 7.50-

7.59 (m, 4H), 7.46 (dd, $J_1 = 8.2$ Hz, $J_2 = 0.8$ Hz, 1H), 7.41 (d, $J = 8.2$ Hz, 2H), 7.26-7.33 (m, 2H),
7.21 (d, $J = 7.6$ Hz, 2H), 7.16 (d, $J = 7.8$ Hz, 1H), 7.05-7.10 (m, 2H), 6.92 (dd, $J_1 = 9.1$ Hz, $J_2 =$
3.7 Hz, 2H), 6.74 (d, $J = 7.8$ Hz, 1H), 6.62 (dd, $J_1 = 5.6$ Hz, $J_2 = 1.8$ Hz, 1H), 3.89 (s, 3H), 3.87
(s, 3H), 2.09 (s, 6H). ^{13}C NMR (126 MHz, CD_2Cl_2 , δ): 163.47, 163.28, 161.26, 160.02, 159.99,
159.19, 159.13, 155.22, 155.15, 151.83, 151.68, 143.87, 143.46, 139.29, 135.05, 135.03, 134.15,
132.70, 132.08, 131.70, 131.62, 131.51, 131.33, 131.30, 130.71, 130.37, 129.71, 129.42, 128.65,
128.50, 128.36, 128.27, 127.83, 127.58, 127.20, 127.13, 127.06, 126.81, 126.80, 126.74, 126.48,
125.84, 125.69, 125.62, 124.38, 124.09, 123.84, 123.55, 123.45, 122.78, 122.37, 121.31, 121.12,
120.75, 119.47, 117.28, 116.81, 116.55, 116.43, 111.00, 110.61, 108.07, 107.94, 28.93, 16.84.
ESI-HRMS (m/z): calculated for $\text{C}_{74}\text{H}_{41}\text{N}_4\text{O}_{12}$ $[\text{M}+\text{H}]^+$: 1177.2715, found: 1177.2707.

PXX-*d*₉-Br. Under a nitrogen atmosphere, **PXX-*d*₁₀** (42) (1 eq, 0.34 mmol, 100 mg) and *N*-
bromosuccinimide (1 eq, 0.34 mmol, 60.9 mg) were suspended in dry dimethylformamide (6.5
mL). The mixture was sparged with nitrogen for 15 minutes, then stirred at room temperature for
16 hr. The mixture was poured into water (60 mL) and vacuum filtered. The yellow solid was
washed three times with water (20 mL), then dried in a vacuum oven overnight to yield **PXX-*d*₉-**
Br as a dull yellow solid (104 mg). The product was used without further purification.

PXX-*d*₉-Bpin. Under a nitrogen atmosphere, **PXX-*d*₉-Br** (1 eq, 0.28 mmol 103 mg),
bis(pinacolato)diboron (3 eq, 0.83 mmol, 212 mg), potassium acetate (6 eq, 1.67 mmol, 164 mg),
and [1,1'-bis(diphenylphosphino)ferrocene]dichloropalladium(II) (0.05 eq, 0.014 mmol, 10.2 mg)
were dissolved in dioxanes (15 mL, stored in an inert atmosphere over molecular sieves). The
mixture was sparged with nitrogen for 15 minutes, then stirred at reflux 12 hr. After cooling to
room temperature, the solvent was removed under reduced pressure. The product was purified with
flash chromatography (50/50 hexanes/dichloromethane mobile phase), yielding **PXX-*d*₉-Bpin** as
a fluorescent yellow solid (50 mg, 43%). ^1H NMR (500 MHz, CDCl_3 , δ): 1.37. ^{13}C NMR (126

MHz, CDCl₃, δ): 155.43, 152.63, 144.22, 143.75, 135.20, 131.25, 121.47, 121.35, 111.91, 111.28, 83.51, 24.98. ESI-HRMS (m/z): calculated for C₂₆H₁₂D₉BO₄ [M]⁺: 417.2098, found: 417.2120.

PXX-*d*₉-xy-Br. Under a nitrogen atmosphere, **PXX-*d*₉-Bpin** (1 eq, 0.083 mmol 34.2 mg), 2,5-dibromo-*p*-xylene (20 eq, 1.64 mmol, 433 mg), potassium carbonate (4 eq, 0.32 mmol, 45.3 mg), and tetrakis(triphenylphosphine)palladium(0) (0.05 eq, 0.004 mmol, 4.8 mg) were dissolved in dioxanes (10 mL, distilled, then stored in an inert atmosphere over molecular sieves) and water (1 mL). The mixture was sparged with nitrogen for 15 minutes, then stirred at reflux 12 hr. After cooling to room temperature, the solvent was removed under reduced pressure. The product was purified with flash chromatography (90/10 hexanes/dichloromethane mobile phase), yielding **PXX-*d*₉-xy-Br** as a fluorescent yellow solid (31 mg, 80%). ¹H NMR (500 MHz, CDCl₃, δ): 7.47 (s, 1H), 7.05 (s, 1H), 2.38 (s, 3H), 2.01 (s, 3H). ¹³C NMR (126 MHz, CDCl₃, δ): 152.58, 152.10, 144.44, 144.18, 138.27, 136.23, 135.15, 133.55, 132.52, 131.38, 131.18, 129.48, 123.83, 121.56, 121.52, 111.76, 111.39, 22.31, 19.14. ESI-HRMS (m/z): calculated for C₂₈H₈D₉BrO₂ [M]⁺: 473.0977, found: 473.0990.

PXX-*d*₉-xy-Bpin. Under a nitrogen atmosphere, **PXX-*d*₉-xy-Br** (1 eq, 0.046 mmol 21.5 mg), bis(pinacolato)diboron (2 eq, 0.092 mmol, 23.4 mg), potassium acetate (4 eq, 0.18 mmol, 18 mg), and [1,1'-bis(diphenylphosphino)ferrocene]dichloropalladium(II) (0.05 eq, 0.0023 mmol, 1.7 mg) were dissolved in dioxanes (5 mL, distilled, then stored in an inert atmosphere over molecular sieves). The mixture was sparged with nitrogen for 15 minutes, then stirred at reflux 12 hr. After cooling to room temperature, the solvent was removed under reduced pressure. The product was purified with flash chromatography (70/30 hexanes/dichloromethane mobile phase), yielding **PXX-*d*₉-xy-Bpin** as a fluorescent yellow solid (18 mg, 75%). ¹H NMR (500 MHz, CDCl₃, δ): 7.69 (s, 1H), 7.00 (s, 1H), 2.52 (s, 3H), 2.03 (s, 3H), 1.36 (s, 12H). ¹³C NMR (126 MHz, CDCl₃, δ): 152.64, 151.90, 144.37, 144.21, 142.28, 141.58, 137.55, 132.94, 132.60, 131.78, 131.17, 129.48,

121.60, 121.46, 111.63, 111.50, 88.26, 83.47, 29.71, 24.93, 24.92, 21.66, 19.11. ESI-HRMS (m/z): calculated for $C_{34}H_{20}D_9BO_4 [M]^+$: 521.2724, found: 521.2755.

Br-NMI-NDI-dtbp. **NIA-dtbp** and **Br-NMI-NH₂** were synthesized as previously reported.^(38, 51) Under a nitrogen atmosphere, Br-NMI-NH₂ (1 eq, 0.25 mmol, 100 mg) and NIA-dtbp (1.5 eq, 0.38 mmol, 173 mg) were dissolved in ethanol (5 mL, 100%). The mixture was sparged with nitrogen for 15 minutes, then stirred at reflux 12 h (overnight). The precipitate was isolated by vacuum filtration while the mixture was still hot, yielding **Br-NMI-NDI-dtbp** as a white powder (142 mg, 67 %). ¹H NMR (500 MHz, CDCl₃, δ): 8.89-8.84 (m, 4H), 8.37 (dd, $J_1 = 7.3$ Hz, $J_2 = 1.2$ Hz, 1H), 8.66 (dd, $J_1 = 8.6$ Hz, $J_2 = 1.2$ Hz, 1H), 8.49 (d, $J = 7.8$ Hz, 1H), 8.11 (d, $J = 7.9$ Hz, 1H), 7.91 (dd, $J_1 = 8.6$ Hz, $J_2 = 7.2$ Hz, 1H), 7.60 (d, $J = 8.7$ Hz, 1H), 7.48 (dd, $J_1 = 8.6$ Hz, $J_2 = 2.3$ Hz, 1H), 7.28-7.21 (m, 2H), 7.00 (dd, $J_1 = 7.3$ Hz, $J_2 = 2.2$ Hz, 1H), 2.21-2.26 (m, 6H), 1.33-1.30 (m, 9H), 1.27 (s, 9H). ¹³C NMR (126 MHz, CDCl₃, δ): 163.84, 163.24, 163.21, 162.42, 150.39, 143.68, 143.66, 135.16, 134.84, 134.79, 134.27, 133.81, 132.59, 131.99, 131.97, 131.75, 131.54, 131.29, 131.23, 130.97, 130.90, 130.88, 129.52, 129.03, 129.01, 128.27, 127.49, 127.46, 127.43, 127.32, 127.29, 127.16, 126.93, 126.73, 123.12, 122.23, 35.57, 34.28, 31.74, 31.19, 17.40. APPI-HRMS (m/z): calculated for $C_{48}H_{38}BrN_3O_6Na [M+Na]^+$: 854.1842, found: 854.1827.

2-d₉. Under a nitrogen atmosphere, **PXX-d₉-xy-Bpin** (1.22 eq, 0.029 mmol 15 mg), **Br-NMI-NDI-dtbp** (1 eq, 0.024 mmol, 19.7 mg), potassium carbonate (2.5 eq, 0.059 mmol, 8.2 mg), and [1,1'-bis(diphenylphosphino)ferrocene]dichloropalladium(II) (0.1 eq, 0.0023 mmol, 1.7 mg) were dissolved in dioxanes (5 mL, distilled, then stored in an inert atmosphere over molecular sieves) and water (0.5 mL, distilled). The mixture was sparged with nitrogen for 15 minutes, then stirred at reflux 12 h (overnight). After cooling to room temperature, the solvent was removed under reduced pressure. The product was purified with flash chromatography (dichloromethane eluent),

yielding **2-d9** as a yellow solid (14 mg, 42%). ¹H NMR (500 MHz, CDCl₃, δ): 8.90-8.81 (m, 4H), 8.78-8.69 (m, 2H), 8.08 (t, J = 8.0 Hz, 1H), 7.80 (t, J = 7.8 Hz, 1H), 7.62 (t, J = 8.4 Hz, 1H), 7.60 (d, J = 8.7 Hz, 1H), 7.48 (dd, J₁ = 8.6 Hz, J₂ = 2.3 Hz, 1H), 7.31 (d, J = 3.0 Hz, 1H), 7.24-7.19 (m, 2H), 7.02 (dd, J₁ = 13.6 Hz, J₂ = 2.2 Hz, 1H), 2.27-2.18 (m, 6H), 2.13 (s, 3H), 2.07 (s, 3H), 1.34-1.31 (m, 9H), 1.28 (s, 9H). ¹³C NMR (126 MHz, CDCl₃, δ): 163.92, 163.86, 163.73, 162.39, 152.60, 152.14, 150.40, 147.28, 147.27, 147.23, 147.21, 144.48, 144.25, 144.23, 143.75, 143.73, 139.53, 139.49, 137.54, 137.48, 135.58, 135.24, 135.22, 134.73, 134.67, 134.45, 134.44, 134.40, 134.39, 134.20, 133.64, 133.62, 133.58, 133.57, 133.25, 133.18, 132.24, 132.08, 132.06, 132.01, 131.96, 131.71, 131.69, 131.61, 131.59, 131.50, 131.48, 131.35, 131.34, 131.32, 131.22, 130.93, 130.87, 130.82, 129.67, 129.59, 129.03, 129.00, 128.97, 128.12, 128.07, 127.57, 127.54, 127.50, 127.45, 127.35, 127.31, 127.17, 127.10, 127.07, 127.05, 126.70, 123.07, 123.04, 121.96, 121.94, 121.91, 121.63, 121.59, 111.90, 111.83, 111.44, 111.40, 35.60, 34.32, 34.30, 31.77, 31.23, 31.21, 29.71, 22.70, 19.69, 19.66, 19.48, 19.47, 17.54, 17.48, 17.47, 17.45, 14.20, 14.12. APPI-HRMS (*m/z*): calculated for C₇₆H₄₆D₉N₃O₈Na [M+Na]⁺: 1169.4452, found: 1169.4423.

2-h9. Under a nitrogen atmosphere, **PXX-xy-Bpin** (1.5 eq, 0.222 mmol, 112.6 mg), **Br-NMI-NDI-dtbp** (1 eq, 0.150 mmol, 124.8 mg), and potassium carbonate (2 eq, 0.300 mmol, 41.5 mg) were dissolved in tetrahydrofuran (15 mL) and water (1.5 mL). The mixture was sparged with nitrogen for 15 minutes. The flask was opened briefly to add tetrakis(triphenylphosphine) palladium(0) (0.05 eq, 0.0075 mmol, 8.7 mg), then sparged with nitrogen for another 15 minutes. The mixture was then heated to reflux and stirred 12 h (overnight). After cooling to room temperature, the solvent was removed under reduced pressure. The product was purified with flash chromatography (using 3% ethyl acetate in dichloromethane as the mobile phase), yielding **2-h9** as a yellow-green solid (100 mg, 59 %). ¹H NMR (500 MHz, CDCl₃, δ): 8.90-8.82 (m, 4H), 8.76 (dt, J₁ = 7.4 Hz, J₂ = 3.0 Hz, 1H), 8.72 (t, J = 7.7 Hz, 1H), 8.07 (t, J = 7.9 Hz, 1H), 7.82-7.74 (m,

2H), 7.60 (d, $J = 8.7$ Hz, 1H), 7.48 (dd, $J_1 = 8.7$ Hz, $J_2 = 2.3$ Hz, 1H), 7.35 (dd, $J_1 = 9.1$ Hz, $J_2 = 1.9$ Hz, 1H), 7.26-7.19 (m, 3H), 7.14-6.96 (m, 6H), 6.93 (dd, $J_1 = 16.6$ Hz, $J_2 = 9.2$ Hz, 1H), 6.78 (dd, $J_1 = 7.8$ Hz, $J_2 = 2.4$ Hz, 1H), 6.67 (m, 1H), 2.27-2.20 (m, 6H), 2.13 (s, 3H), 2.07 (s, 3H), 1.34-1.32 (m, 9H), 1.28 (s, 9H). ^{13}C NMR (126 MHz, CDCl_3 , δ): 163.93, 163.92, 163.86, 163.74, 163.73, 162.40, 152.65, 152.64, 152.18, 150.40, 147.28, 147.24, 144.54, 144.54, 144.31, 144.29, 143.75, 143.74, 139.55, 139.51, 137.54, 137.49, 135.58, 135.58, 135.26, 135.23, 134.74, 134.69, 134.47, 134.45, 134.41, 134.40, 134.20, 133.64, 133.64, 133.59, 133.58, 133.28, 133.27, 133.19, 133.18, 132.27, 132.25, 132.15, 132.10, 132.06, 131.73, 131.72, 131.61, 131.58, 131.57, 131.51, 131.49, 131.46, 131.38, 131.36, 131.35, 131.33, 130.94, 130.88, 130.82, 129.75, 129.68, 129.04, 129.01, 128.98, 128.12, 128.08, 127.84, 127.72, 127.57, 127.54, 127.50, 127.45, 127.35, 127.31, 127.24, 127.18, 127.10, 127.06, 126.70, 126.61, 126.58, 125.31, 125.21, 123.07, 123.03, 121.96, 121.93, 121.91, 121.63, 121.59, 120.26, 120.24, 117.52, 117.39, 117.38, 111.92, 111.84, 111.45, 111.42, 108.78, 108.64, 108.56, 35.60, 34.32, 34.30, 31.77, 31.23, 31.22, 29.71, 19.70, 19.66, 19.48, 19.47, 17.54, 17.48, 17.47, 17.45. APPI-HRMS (m/z): calculated for $\text{C}_{76}\text{H}_{55}\text{N}_3\text{O}_8$ ESI-HRMS (m/z): calculated for $\text{C}_{76}\text{H}_{55}\text{N}_3\text{O}_8$ $[\text{M}]^+$: 1137.3984, found: 1137.3998.

Optical Spectroscopic Characterization

The two enantiomers of (*R,S*)-**1-h**₉ and (*R,S*)-**1-d**₉ were separated using a chiral HPLC column (Fig. S3). Their steady-state UV-Vis spectra were obtained with a Shimadzu-1800 spectrophotometer in toluene. The spectrum of racemic (*R,S*)-**1-h**₉ along with the spectra of PXX-**h**₉, racemic (*R,S*)-NMI₂, and NDI are shown in Fig. S4a. The spectrum of (*R,S*)-**1-h**₉ is approximately an additive combination of those of PXX, (*R,S*)-NMI₂, and NDI, thus indicating that the electronic coupling between these redox components is relatively weak. The corresponding spectra of (*R,S*)-**1-d**₉ are nearly identical (Fig. S4c). Circular dichroism (CD) spectra of each enantiomer show the expected mirror image symmetry over the absorption range of the chiral

NMI₂ bridge molecule (Figs. S4b and S4d), which is in full agreement with the CD spectra reported for NMI₂ previously.⁽³²⁾

Details of the transient absorption instrumentation have been described previously.⁽⁵²⁾ Briefly, ~50% of the output of a 1 kHz amplified Ti:sapphire system at 827 nm (1 W, 100 fs, Spitfire, Spectra Physics) is used to pump a non-collinear optical parametric amplifier (TOPAS-Prime, Light-Conversion, LLC.) tuned to generate the ~120 fs, 520 nm pump pulses. The pump is depolarized to minimize polarization-specific dynamics. In the nsTA experiment, the probe is generated in a separately delayed broadband laser system (EOS, Ultrafast Systems, LLC). The transmitted probe is detected on a commercial spectrometer (customized Helios-EOS, Ultrafast Systems, LLC).

Transient absorption (TA) measurements were used to obtain the photoinduced electron transfer dynamics for the racemic mixtures (*R* and *S*)-**1-h₉**, (*R* and *S*)-**1-d₉** as well as the achiral reference molecules **2-h₉** and **2-d₉** in a butyronitrile (PrCN) glass at 105 K using a 120-fs, 450 nm pump pulse and monitoring the transient spectra at 100 fs-250 μs (Fig. S5). These data were globally fit over the entire time range to yield evolution-associated spectra (Fig. S6). Photoexcitation of (*R* and *S*)-**1-h₉** results in the ground state bleach (GSB) of its 450 nm absorption accompanied by the appearance of the excited state absorption (ESA) bands of the lowest excited singlet state of PXX-*h₉* (¹*PXX) at 630, 779, and 1177 nm, as well as stimulated emission features at 483 and 523 nm (Figs. 5a and S6a). These transient features decay in $\tau_{CS1} = 12.4 \pm 0.4$ ps accompanied by the rise of a complex transient signal showing a prominent 979 nm band characteristic of PXX-*h₉*^{•+}.⁽³¹⁾ The 410 nm absorption characteristic of NMI[•] is largely cancelled out by the PXX-*h₉* GSB.⁽⁵³⁾ A second electron transfer producing NDI[•] occurs in $\tau_{CS2} = 203 \pm 8$ ps as indicated by the formation of the sharp features at 474 and 601 nm.⁽⁵³⁾ Further low amplitude absorption changes of PXX-*h₉*^{•+} and NDI[•] occur in $\tau_{rlx} = 32 \pm 5$ ns, which are attributed to a

relaxation process in the low temperature glassy solvent followed by decay of the PXX-*h*₉^{•+}-NDI^{•-} SCRPs to its ground state in $\tau_{CR} = 65.9 \pm 0.7 \mu s$. The corresponding TA data for (*R* and *S*)-**1-d**₉ (Figs. S5c and S6c) exhibit spectra and kinetics very similar to those of (*R* and *S*)-**1-h**₉, where the charge recombination time of the PXX-*d*₉^{•+}-NDI^{•-} SCRPs to its ground state $\tau_{CR} = 51.1 \pm 0.3 \mu s$.

TA measurements under the same conditions were conducted on the achiral reference molecule **2-h**₉. The transient spectra at selected times are shown in Fig. S5b. The spectral features are very similar to those of (*R* and *S*)-**1-h**₉, and the time constants for charge separation and recombination obtained from the global fits to the data are $\tau_{CS1} = 4.8 \pm 0.3$ ps, $\tau_{CS2} = 213 \pm 6$ ps, $\tau_{rlx} = 4.8 \pm 0.5$ ns, and $\tau_{CR} = 46.4 \pm 0.6 \mu s$. The TA data for achiral **2-d**₉ (Figs. S5d and S6d) exhibit spectra and kinetics very similar to those of **1-h**₉, where the charge recombination time of the PXX-*d*₉^{•+}-NDI^{•-} SCRPs to its ground state in $\tau_{CR} = 57.0 \pm 0.1 \mu s$. The TA data show that (*R* and *S*)-**1** and **2** have ultrafast charge separation rates, which ensures an initial singlet state precursor, as well as long SCRPs recombination lifetimes, making them excellent candidates for studying the CISS effect on their spin dynamics.

Transient Electron Paramagnetic Resonance (TREPR) Spectroscopy

All EPR measurements were performed at X-band (~9.6 GHz) on a Bruker Elexsys E680 X/W EPR spectrometer equipped with a split ring resonator (ER4118X-MS3). The temperature was maintained at 85 K using an Oxford Instruments CF935 continuous-flow cryostat using liquid nitrogen. The sample was photoexcited at 450 nm with 7 ns, 1 mJ laser pulses generated via an optical parametric oscillator (Spectra-Physics BasiScan) pumped with the 355 nm output of a frequency-tripled Nd:YAG laser (Spectra-Physics Quanta-Ray Lab-170+-10H) operating at a repetition rate of 10 Hz. The laser light was coupled into the resonator via a fiber optic (Thorlabs FT1000UMT) and collimator placed outside the cryostat window.

All EPR measurements were performed in the liquid crystal, 5CB (Sigma-Aldrich). Solutions with an O.D. of 0.5 over a 2 mm pathlength were loaded into quartz tubes (2.40 mm o.d., 2.00 mm i.d.), subjected to three freeze-pump-thaw cycles on a vacuum line (10^{-3} Torr), and sealed with a hydrogen torch. The samples slightly warmed, then aligned in a 1 T magnetic field for 5 minutes and quickly loaded into the already cold resonator at 85 K. Two sample orientations were collected, one, as loaded, with the 5CB director parallel to the magnetic field, and other by rotating the sample 90 degrees to generate a liquid crystal director perpendicular to the magnetic field.

Transient continuous-wave EPR (TREPR) spectra were collected following photoexcitation, the kinetic traces of the transient magnetization at a fixed magnetic field were acquired in quadrature under CW irradiation (0.62 mW). Sweeping the magnetic field gave 2D spectra of magnetic field versus time. For each kinetic trace, the average of the signal acquired prior to the laser pulse was subtracted from the data. The kinetic traces recorded at magnetic field values off-resonance were considered background signals, whose average was also subtracted from all kinetic traces. The spectra were then smoothed in time domain using a Savitzky-Golay filter and appropriately phased into absorption/emission and dispersion. All processing and fitting of the spectra were performed in Matlab using home written scripts and the simulation package EasySpin v6.0-dev.(54, 55)

Out-of-phase electron spin echo envelope modulation (OOP-ESEEM) spectroscopy was collected 50 ns following photoexcitation using a two-pulse, $\pi/2 - \tau - \pi$, pulse sequence. Microwave pulses were generated using a 1 kW TWT amplifier (Applied Systems Engineering 117X) in a partially over-coupled resonator to minimize ringing. Using pulse lengths of $\pi/2 = 16$ ns and $\pi = 32$ ns, the pulse sequence was started at $\tau = 100$ ns and incremented $\Delta\tau = 16$ ns, a standard 4-step phase cycle was utilized and the integrated spin echo intensity was recorded.

The time domain OOP-ESEEM traces were fit with the equation:(38)

$$S(\tau) = e^{-\tau/T_d} \int_0^\pi \sin(J + D(\theta)) \tau W(\theta) \sin \theta d\theta \quad (S1)$$

where T_d is a relaxation time, J is the exchange coupling and $D(\theta)$ is the dipolar coupling and $W(\theta)$ is the weighting function ordering imposed by the liquid crystal. The dipolar coupling takes the form:

$$D(\theta) = \frac{\mu_0}{4\pi h} \frac{g_D g_A \beta_e^2}{r_{DA}^3} (1 - 3\cos^2 \theta) \quad (S2)$$

When calculating the distance, g_D and g_A were both taken as the free electron g-value, $g_e = 2.0023$. In the main text we have also introduced the constant $d \equiv \frac{\mu_0}{4\pi h} \frac{g_D g_A \beta_e^2}{r_{DA}^3} = 52.04 \text{ MHz nm}^3 / r_{DA}^3$. The parallel and perpendicular spectra were fit simultaneously with weighting function $W(\theta) = e^{\lambda(3\cos^2 \theta - 1)/2}$, where $\lambda = 5$ for the parallel spectra and $\lambda = -5$ for the perpendicular spectra, resulting in an orientation distribution with a FWHM of $\sim 20^\circ$. We have checked that this orientation distribution does not produce a visible effect on the resulting TREPR spectra.

The OOP-ESEEM data and fits are shown in Figs. S7 and S8 and the fitting parameters are summarized in Table S1.

Density Functional Theory Calculations

Geometry optimizations of (*R*)-**1-h**, (*S*)-**1-h**, and **2-h** were performed using density functional theory (DFT) at B3LYP/6-31G* level of theory in QChem (version 5.1).⁽⁵⁶⁾ The optimized structures are shown in Fig. S9 and the optimized cartesian coordinates are given in Tables S2-S4.

Supplementary Text

Spin-Correlated Radical Pair (SCRCP) Theory

SCRCP with no CISS Effect

The radical pair spin Hamiltonian is:

$$\hat{\mathcal{H}} = \beta_e \mathbf{B}_0 g_D \hat{\mathbf{S}}_D + \beta_e \mathbf{B}_0 g_A \hat{\mathbf{S}}_A + J \hat{\mathbf{S}}_D \hat{\mathbf{S}}_A + \hat{\mathbf{S}}_D \mathbf{D}_{DA} \hat{\mathbf{S}}_A + \hat{\mathcal{H}}_{HF} \quad (S3)$$

The first two terms are the electron Zeeman interaction with the static magnetic field, $\mathbf{B}_0$; J is the exchange coupling; $\mathbf{D}_{DA}$ is the electron-electron dipole interaction matrix with principal elements of $[D \quad D \quad -2D]$ and D follows the definition given in eqn S2; and within the final term, $\hat{\mathcal{H}}_{HF}$, are the electron-nuclear hyperfine interactions.

5 In the coupled spin basis, $|T_+\rangle, |T_0\rangle, |T_-\rangle, |S\rangle$, this leads to the Hamiltonian matrix:(39, 40, 57)

$$\hat{\mathcal{H}} = \begin{bmatrix} v_+ + \frac{J}{4} + \frac{D}{4} & 0 & 0 & 0 \\ 0 & \frac{J}{4} - \frac{D}{2} & 0 & v_- \\ 0 & 0 & -v_+ + \frac{J}{4} + \frac{D}{4} & 0 \\ 0 & v_- & 0 & -\frac{3J}{4} \end{bmatrix} \quad (\text{S4})$$

where:

$$v_+ = \frac{1}{2}\beta_e(g_D + g_A)B_0 + \sum_j(a_{Dj} + a_{Aj})m_j \quad (\text{S5a})$$

$$v_- = \frac{1}{2}\beta_e(g_D - g_A)B_0 + \sum_j(a_{Dj} - a_{Aj})m_j \quad (\text{S5b})$$

10 The angle θ follows the standard definition, the angle between the vector connecting the two spins and the laboratory magnetic field. More generally, the Hamiltonian matrix, eq. S4, is written in the laboratory frame as defined by the high-field approximation. It is readily apparent that the off-diagonal term, v_- , (i.e. the difference between the effective Zeeman energies of the two ions, also including the contribution from hyperfine couplings) will mix the $|T_0\rangle$ and $|S\rangle$ states and upon
15 diagonalization yields the eigenstates:

$$\begin{aligned} |1\rangle &= |T_{+1}\rangle \\ |2\rangle &\equiv |\Phi_A\rangle = \cos\phi |S\rangle + \sin\phi |T_0\rangle \\ |3\rangle &\equiv |\Phi_B\rangle = -\sin\phi |S\rangle + \cos\phi |T_0\rangle \\ |4\rangle &= |T_{-1}\rangle \end{aligned} \quad (\text{S6})$$

when

$$\phi = \frac{1}{2}\tan^{-1} \frac{2v_-}{(D/2)-J} \quad (\text{S7})$$

And the eigenvalues:

$$\begin{aligned}
 E_1 &= \nu_+ + \frac{(J+D)}{4} \\
 E_2 &= -\frac{(J+D)}{4} + \sqrt{\left(\frac{D}{4} - \frac{J}{2}\right)^2 + \nu_-^2} \\
 E_3 &= -\frac{(J+D)}{4} - \sqrt{\left(\frac{D}{4} - \frac{J}{2}\right)^2 + \nu_-^2} \\
 E_4 &= -\nu_+ + \frac{(J+D)}{4}
 \end{aligned} \tag{S8}$$

The spin Hamiltonian and its resulting eigenvalues and eigenvectors are independent of any molecular chirality and only depend on the magnetic properties of the involved spins.

The steady-state EPR spectrum (after coherences are lost) in field or frequency space is composed of peaks at positions determined by differences in the eigenvalues of the spin Hamiltonian, eqns. S8, with intensities determined by the transition rate between the eigenstates $|i\rangle$ and $|j\rangle$ and their population differences:

$$I_{ij} \propto |\langle i | \hat{S}_y | j \rangle|^2 (p_i - p_j) \tag{S9}$$

where p_i and p_j are the populations of the two eigenstates which correspond to the diagonal elements of the density matrix in the eigenbasis and depend upon on the initial state, i.e. $p_i = |\langle i | \psi_0 \rangle|^2$. $\langle i | \hat{S}_y | j \rangle$ is the magnetic dipole matrix element of the examined transition along the direction of the microwave field, which is orthogonal to the static field.

Conversely, transition frequencies (and hence peak positions) only depend on the Hamiltonian eigenvalues and not on the specific initial state.

The four transitions of the SCRP (starting from a singlet precursor) appear at their usual positions.

All four transitions are of the same intensity (see definition in Eq. S9) such that:

$$I_{12} = -I_{24} = I_{13} = -I_{34} = \frac{\cos^2 \phi \sin^2 \phi}{2} = \frac{\nu_-^2}{2(D/2 - J)^2 + 8\nu_-^2} \tag{S10}$$

SCRP with the CISS Effect

The state for a radical pair following electron transfer through a chiral bridge can be written as

$$|\psi_0\rangle = \cos \chi |S\rangle + \sin \chi |T_0(\mathbf{n})\rangle \tag{S11}$$

This state was first proposed by Luo and Hore,(46) where the angle χ is related to the chirality and quantifies the effect of CISS in the reaction. The states $|S\rangle$ and $|T_0(\mathbf{n})\rangle$ are the radical pair singlet and the radical pair triplet states, respectively; in particular, the triplet state is defined with respect to the axis $\mathbf{n}$ of charge separation, which is also the axis of chirality, the CISS quantization axis, and the long molecular axis. In the limiting cases: $\chi = 0$ (no CISS), $|\psi_0\rangle$ is a pure singlet state; at the other extreme $\chi = \pm \frac{\pi}{4}$, so that $|\psi_0\rangle = |\uparrow\downarrow(\mathbf{n})\rangle$ or $|\downarrow\uparrow(\mathbf{n})\rangle$, i.e. it shows maximum local spin polarization (100% CISS); and when $\chi = \pm \frac{\pi}{2}$, $|\psi_0\rangle = |T_0(\mathbf{n})\rangle$ a pure triplet state. The $\pm$ refers to the two different enantiomers.

In the basis of the spin Hamiltonian, eqn. S4, the triplet state is defined in the high-field limit and polarized along the laboratory magnetic field. When working in orientations or distributions of orientations where the CISS axis is not aligned with the laboratory magnetic field, it becomes necessary to rotate the state as defined in the molecular frame to the laboratory frame:

$$e^{-i\hat{S}_y\theta}|\psi_0\rangle = |\psi_0(\theta)\rangle = \cos\chi|S\rangle + \sin\chi\left[\cos\theta|T_0\rangle + \frac{\sin\theta}{\sqrt{2}}(|T_{-1}\rangle - |T_{+1}\rangle)\right] \quad (\text{S12})$$

where the singlet component is rotationally invariant, while $|T_0(\mathbf{n})\rangle \equiv |T_0(\theta)\rangle = \cos\theta|T_0\rangle + \frac{\sin\theta}{\sqrt{2}}(|T_{-1}\rangle - |T_{+1}\rangle)$. Similar to eqn. S2, θ is defined as the angle between the CISS axis and the magnetic field. In the $D^{*+}\text{-}B\chi\text{-}A^{*-}$ molecules studied here, the CISS axis and the dipolar axis are nearly collinear so θ in eqns. S2 and S12 refers to the same measure. When $\theta \neq 0$, sizable components on $|T_{-1}\rangle$ and $|T_{+1}\rangle$ are introduced (see Fig. 3 of the main text).

Let us now focus on the two limiting cases examined in this work, i.e. $\theta = 0^\circ$ and $\theta = 90^\circ$.

The resulting intensity for I_{13} and I_{34} at $\theta = 90^\circ$ is given by:

$$I_{13} = -I_{34} = \frac{1}{4}\cos^2\phi(\cos 2\chi - \cos^2\chi\cos 2\phi) \quad (\text{S13})$$

Showing a dependence on the composition of the eigenstates (ϕ) and on the amount of CISS (χ).

This can be separated into a singlet (I_S) and a 100% CISS (I_{CISS}) contributions, i.e.

$$\begin{aligned} I_{13} = -I_{34} &= \frac{1}{2}(1 - 2 \sin^2 \chi) \cos^2 \phi \sin^2 \phi - \frac{1}{4} \sin^2 \chi \cos^2 \phi \cos 2\phi \\ &= (1 - p) I_S + p I_{CISS} \end{aligned} \quad (S14)$$

where $p = 2 \sin^2 \chi$ represents the CISS efficiency and

$$I_S = \frac{1}{2} \cos^2 \phi \sin^2 \phi \quad (S15)$$

$$I_{CISS} = -\frac{1}{8} \cos 2\phi \cos^2 \phi \quad (S16)$$

Note that for this orientation I_S and I_{CISS} have opposite signs and that I_{CISS} is the same that would be obtained starting from a fully polarized initial state ($\chi = \pm\pi/4$).

The calculation is perfectly analogous for $I_{12} = -I_{24}$, so that

$$I_{12} = -I_{24} = \frac{1}{4} \sin^2 \phi (2 \cos^2 \phi \cos^2 \chi - \sin^2 \chi) \quad (S17)$$

This can also be decomposed into singlet and 100% CISS contributions, with the same sign. Remarkably, in the present case I_{12} and I_{24} are much weaker and occur in the central part of the spectrum. Hence, we focus on the main peaks I_{34} and I_{13} in the following. Similar expressions can also be derived for $\theta = 0$. A summary of the different singlet and CISS contributions for each orientation is reported in Table S5.

In Table S5, blue (orange) indicate emissive (absorptive) signals, fixed by the sign of D . Note that opposite enantiomers are indistinguishable when $\theta = 90^\circ$, while they give different intensities when $\theta = 0^\circ$, which are therefore averaged in the Table S5.

For these specific directions the same results could be obtained using the initial mixed state for electron transfer through a chiral bridge proposed by Chiesa et al.(47) or the coherent superposition

$$|\psi'_0\rangle = \cos \chi |S\rangle + i \sin \chi |T_0(\mathbf{n})\rangle \quad (S18)$$

derived by Fay.(45) Eq. S18 does not yield any single-spin polarization, but was expanded later by Fay(44) to account for spin-spin exchange interactions during multi-step electron transfer processes, leading to polarization effects similar to those postulated by Luo and Hore(46) (a mixture of states S18 and S11 with the same angle χ). Since the three models are equivalent for describing the TREPR spectra, we mainly focus on Eq. S11.(46)

TREPR Spectral Linewidths

To understand how the entanglement of the eigenstates affects the linewidths for different initial states, let us focus again on $\theta = 90^\circ$ and transition I_{13} between $|\Phi_B\rangle$ and $|T_{+1}\rangle$, which occurs on the low field side of the spectrum. Analogous reasoning holds for transition I_{34} on the high field side of the spectrum. The intensities of the transitions for singlet and 100% CISS initial states are given by Eqs. S15 and S16 above. Their dependence on the mixing angle ϕ is very different, as shown in Fig. S10. In order to better compare the linewidths of the two curves, here we report the absolute value of I_S and I_{CISS} (Fig. S10A), along with its derivative (Fig. S10B).

As discussed in the main text, ϕ can be varied by considering different configurations of the nuclear spins coupled to the SCRP or by distributions of the Hamiltonian parameters. In the present calculations, we included a 14% strain in the donor-acceptor distance, leading to a distribution of dipole-dipole interactions. This, in turn, corresponds to resonances occurring at different values of B_0 . As a result, we can have regions in the tails of a specific peak, which correspond to more entangled states, i.e. such that ϕ is closer to 0 compared to the center of the peak. In this case, moving from the center of the peak to the tail corresponds to moving toward $\phi = 0$ in Fig. S10B, where $d|I_{CISS}|/d\phi$ is positive and $d|I_S|/d\phi$ is negative. Such a situation, in which I_{CISS} increases while I_S decreases from the center to the tail of the peak leads to a larger broadening of I_{CISS} compared to I_S , thus exactly producing CISS features in the tail when I_{CISS} and I_S are combined.

TREPR Spectral Simulations

The time dependence of the TREPR spectra is simulated by numerically integrating the stochastic Liouville equation for the system density matrix

$$\frac{d\rho_r(\Omega, t)}{dt} = -i \left[\frac{1}{\hbar} \tilde{H}(\Omega) + i\tilde{R} + i\tilde{K} \right] \rho_r(\Omega, t) \quad (\text{S19})$$

where $\rho_r(\Omega, t)$ is the density matrix of the system in the rotating frame, $\tilde{H}(\Omega)$, $\tilde{R}$ and $\tilde{K}$ are super-operators associated to the coherent Hamiltonian evolution, to dephasing/relaxation and recombination effects, respectively. Both relaxation and recombination are negligible in the time range of interest and we therefore omit them in the following. Conversely, off-diagonal elements of ρ_r are affected by pure dephasing, parametrized by transition-dependent T_2 .

Note that in the high-field limit, the non-secular terms in the Hamiltonian are also negligible and the rotating-frame Hamiltonian (including the microwave field) becomes time independent. Hence, Eq. S19 can be directly integrated, leading to

$$\rho_r(\Omega, t) = \exp \left[-\frac{i}{\hbar} \tilde{H}(\Omega) + \tilde{R} + \tilde{K} \right] t \rho_r(\Omega, 0) \quad (\text{S20})$$

from which the spectrum $\langle S_y(B_0, t) \rangle = \text{Tr}[\rho_r(\Omega, t) S_y]$ is calculated.(47)

Apart from the rise of the signal at short times, the resulting spectra are equivalent to those obtained from linear-response theory in the stationary limit, Eq. S9, using, for instance, the Easyspin package for spin-polarized initial states.(54, 55)

The time and field dependencies of the TREPR spectra simulated for the protonated compound using the minimal model described in the main text and simulated for the deuterated case including a distribution of hyperfine fields generated by the four ^1H and two ^{14}N nuclear spins of $\text{NDI}^\bullet$ are reported in Figs. S11 and S12. The experimental spectra are very well reproduced, both for the main peaks and for the characteristic CISS features at the two wings. The latter show practically the same time dependence as the main peaks, indicating that they are not transient but resonant

peaks (also present in the steady-state spectra). The full distributions of hyperfine fields, g-tensors and dipolar strain included in this model are slightly different from those reported in the main text, but these small differences do not affect the origin of CISS features.

Electron Transfer Rates and Energetics

The free energies of reaction for (*R* or *S*) **1-h9** (**-d9**) and **2-h9** (**-d9**) were determined from the modified Weller equation(58) (Eq. S21) using the one-electron oxidation potential $E_{1/2}^+$ of PXX (0.81 V vs. SCE(31), and the one-electron reduction potentials $E_{1/2}^-$ of NMI, NMI₂, and NDI (-1.4 V, (53) -1.5 V(59) and -0.48 V (53) vs SCE, respectively), the estimated dielectric constant of PrCN at 105 K ($\epsilon = 2.3$),(60) and the center-to-center radical ion pair distances given in Table S1 determined as noted. The ionic radii are estimated as half of the donor-acceptor distance and e_0 is the charge of the electron.

$$\Delta G_{DA} = E_{1/2}^+ - E_{1/2}^- + \frac{e_0^2}{\epsilon} \left(\frac{1}{2r_D} + \frac{1}{2r_A} - \frac{1}{r_{DA}} \right) \quad (\text{S21})$$

The values of the electronic coupling matrix elements for the electron transfer reaction were determined using the semi-classical expression for the electron transfer rate k_{ET} given in Eq. S22, (60) where ΔG_{DA} and k_{ET} are the free energies of reaction and measured rate constants given in Table S1. The values of k_{ET} are obtained from the data presented in Figs. S5 and S6. In addition, the total internal reorganization for the one-electron oxidation of PXX and reduction of NMI, NMI₂ or NDI (λ_i) is estimated as 0.5 eV, while the solvent reorganization energy of PrCN at 105 K (λ_s) is estimated from Eq. S23(60) and we make the usual assumption of one dominant vibration ($\omega = 1500 \text{ cm}^{-1}$) coupled to the reaction coordinate of the electron transfer reaction.

$$k_{ET} = \frac{2\pi}{\hbar} V_{DA}^2 \frac{1}{(4\pi\lambda_s k_B T)^{1/2}} \sum_{m=1}^{\infty} \frac{s^m e^{-s}}{m!} \cdot \exp \left[-\frac{(\lambda_s + \Delta G_{DA} + m\omega)^2}{4\lambda_s k_B T} \right] \quad (\text{S22})$$

where $s = \lambda_i/\omega$

$$\lambda_s = e_0^2 \left(\frac{1}{2r_D} + \frac{1}{2r_A} - \frac{1}{r_{DA}} \right) \left(\frac{1}{n^2} - \frac{1}{\epsilon} \right) \quad (\text{S23})$$

where n is the refractive index of the solvent and the remaining values are defined above.

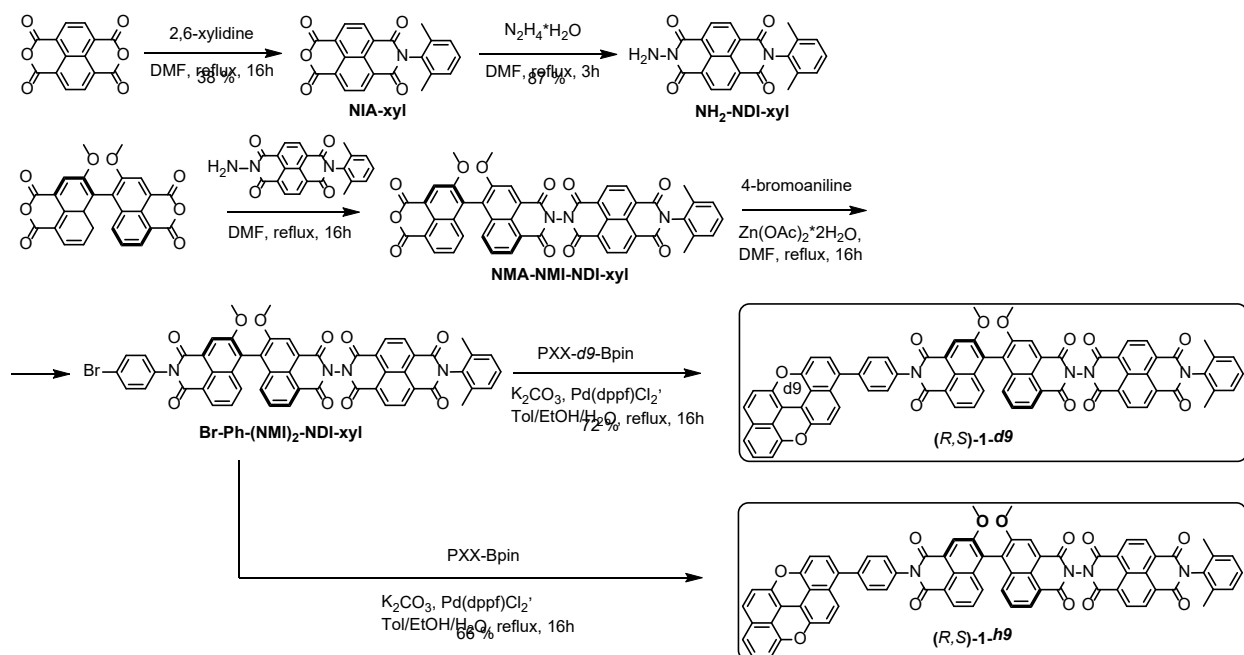


Fig. S1. Synthetic scheme for (*R* and *S*)-1-*h*₉ and (*R* and *S*)-1-*d*₉

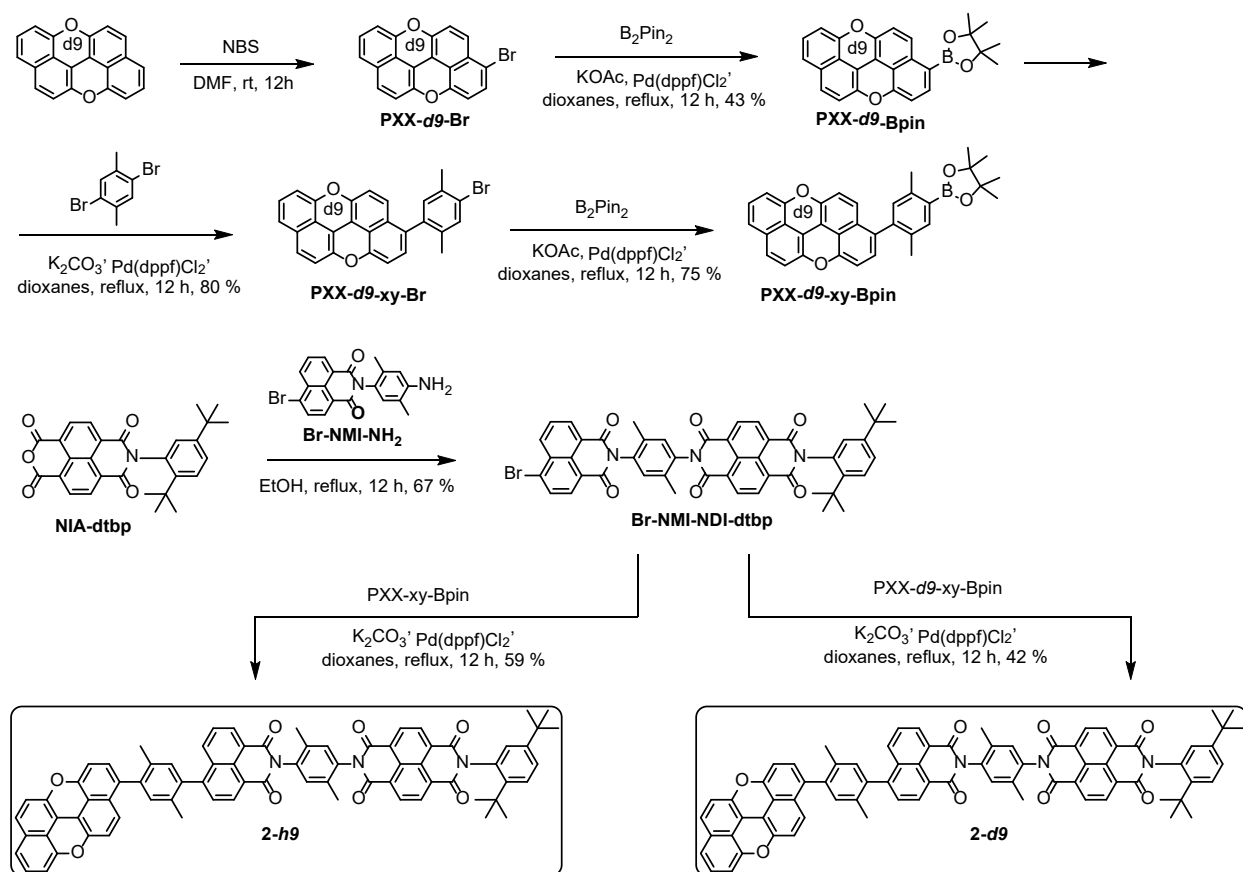


Fig. S2. Synthetic scheme for 2-h9 and 2-d9

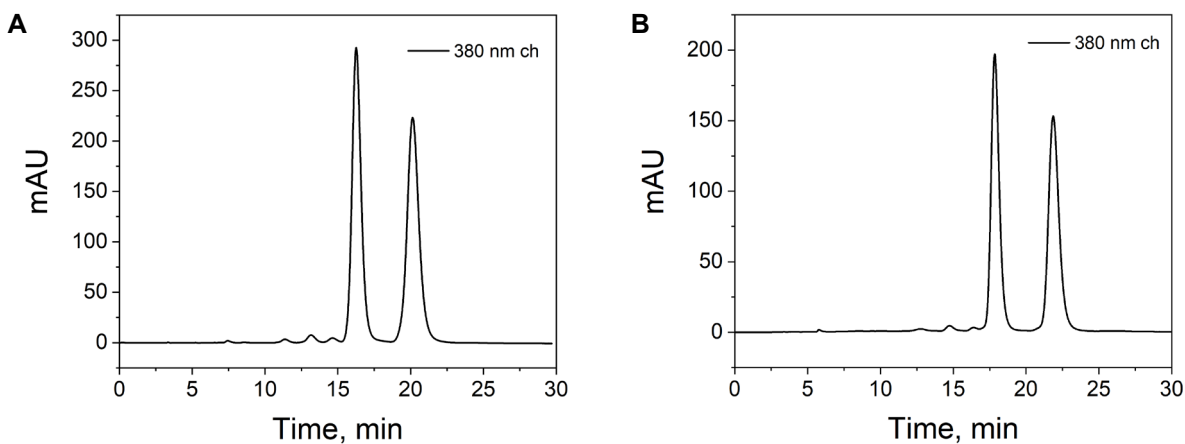


Fig. S3. HPLC chromatogram showing the chiral resolution of (A) (*R* and *S*)-1-*h*₉ and (B) (*R* and *S*)-1-*d*₉ enantiomers on a ChiralPak IA-3 column (250 x 4.6 mm, 3 μm) monitored at 380 nm using a dichloromethane/hexanes/isopropanol mixture as an eluent (50/45/5 volume %) at 1 mL/min flow rate. Peaks with the shorter retention times are the (*S*) enantiomers.

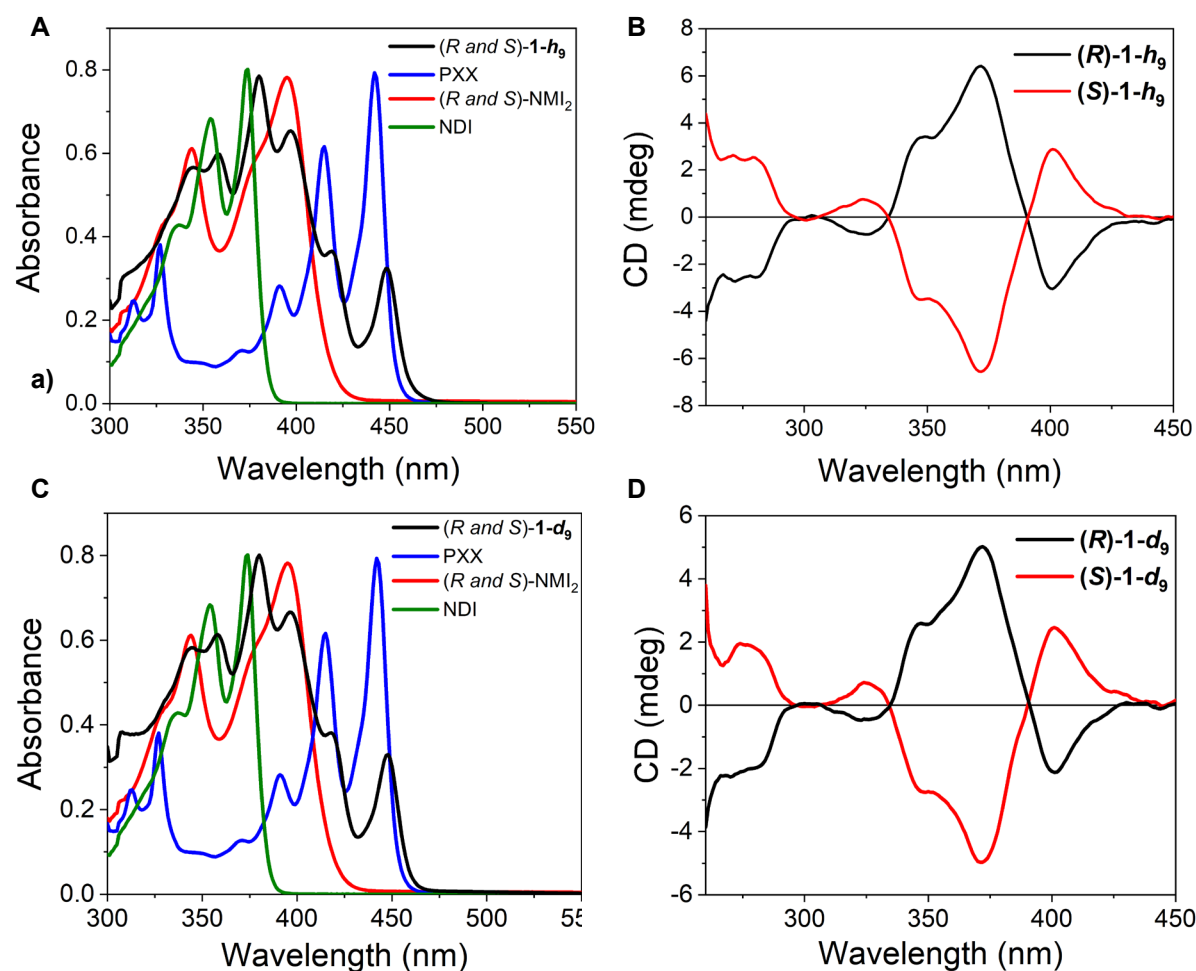


Fig. S4. Optical spectra. (A) UV-vis absorption of the racemic mixture of $(R$ and S)- $1-h_9$, PXX, $(R$ and S)- NMI_2 and NDI in toluene at 295 K. (B) CD spectra of (R) - $1-h_9$ and (S) - $1-h_9$. (C) UV-vis absorption of the racemic mixture of $(R$ and S)- $1-d_9$, PXX, $(R$ and S)- NMI_2 and NDI in toluene at 295 K. (D) CD spectra of (R) - $1-d_9$ and (S) - $1-d_9$.

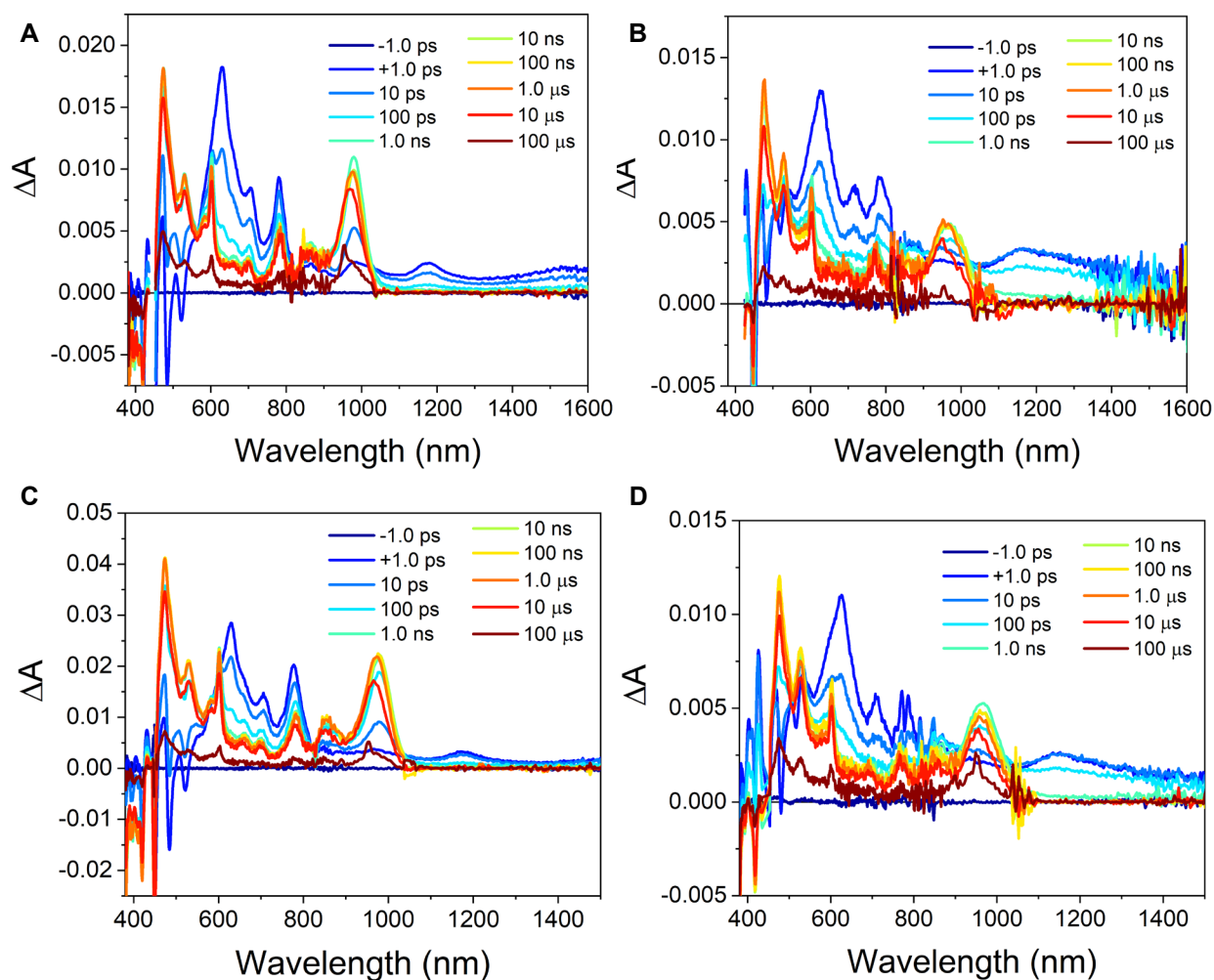


Fig. S5. Transient absorption data following 450 nm excitation of compounds in PrCN at 105 K at the indicated pump-probe delay times. (A) (*R* and *S*)-**1-h**, (B) **2-h**, (C) (*R* and *S*)-**1-d**, (D) **2-d**.

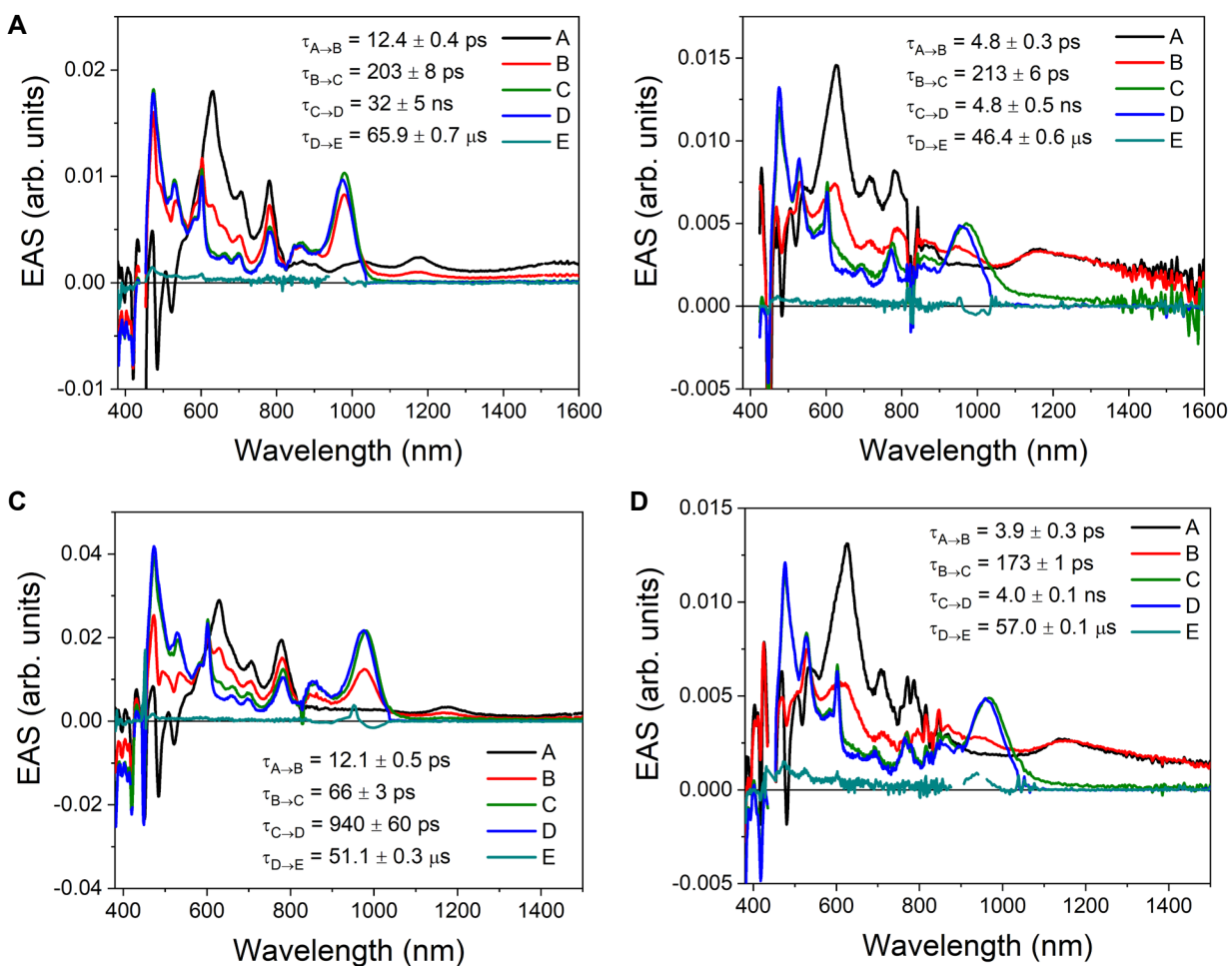


Fig. S6. Evolution-associated spectra obtained by globally fitting the femto- and nanosecond transient absorption data for (A) (*R* and *S*)-**1-h**, (B) **2-h**, (C) (*R* and *S*)-**1-d**, and (D) **2-d**, following 450 nm excitation in PrCN at 105 K.

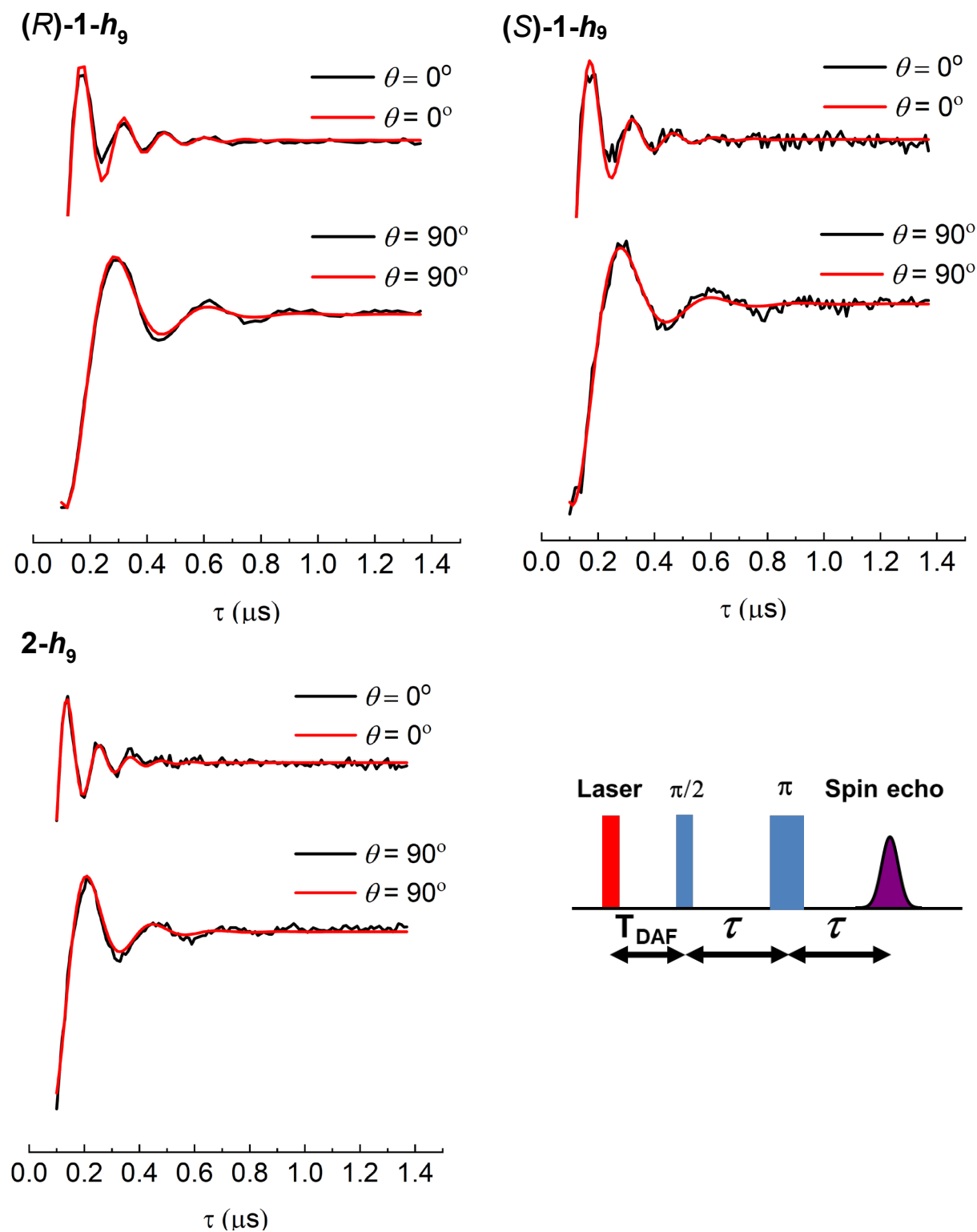


Fig. S7. OOP-ESEEM data and fits at $\theta = 0^\circ$ and $\theta = 90^\circ$ for the indicated molecules oriented in 5CB. Lower right: Hahn echo pulse sequence for OOP-ESEEM experiments, where T_{DAF} is the delay after the laser flash.

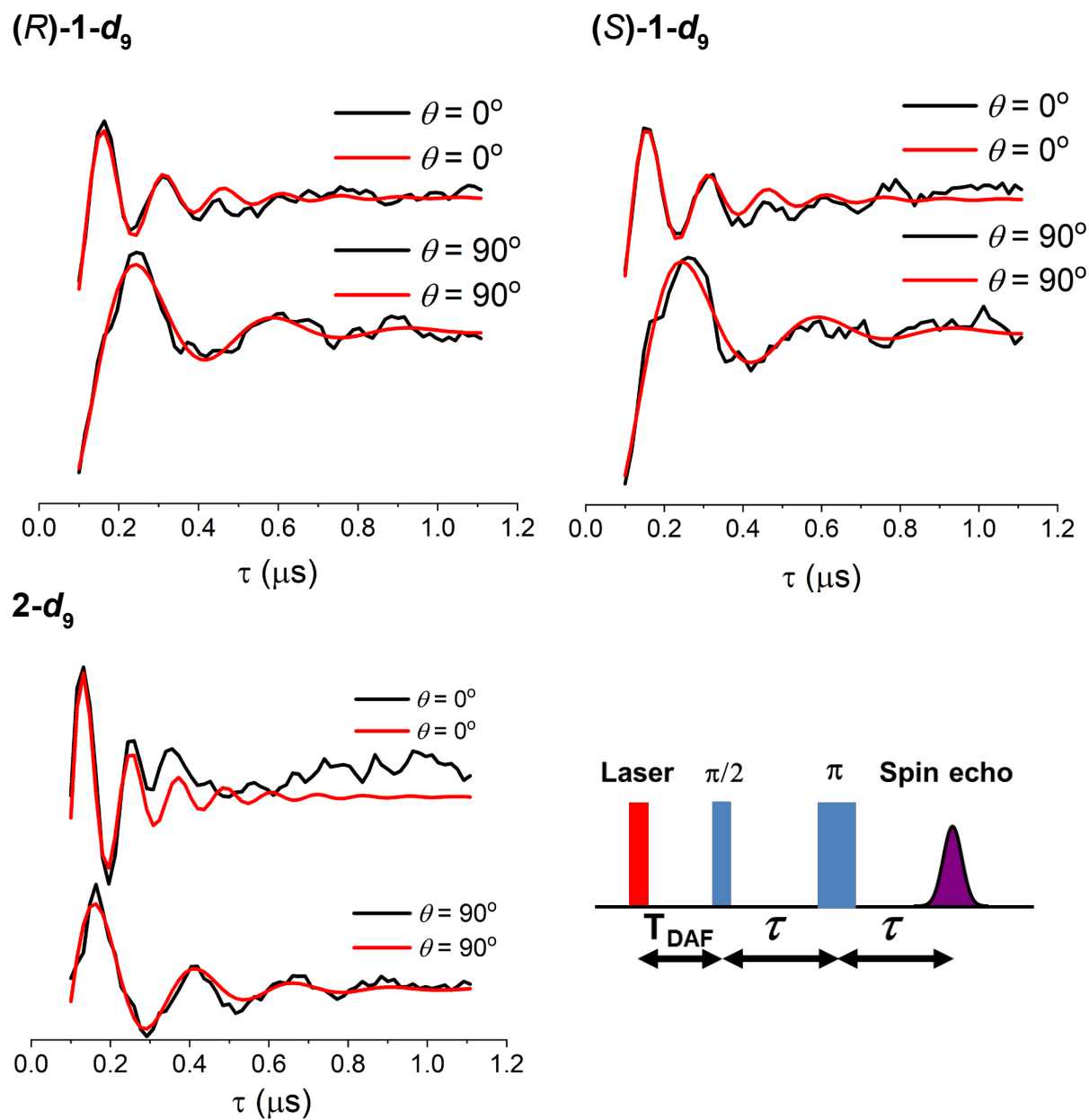
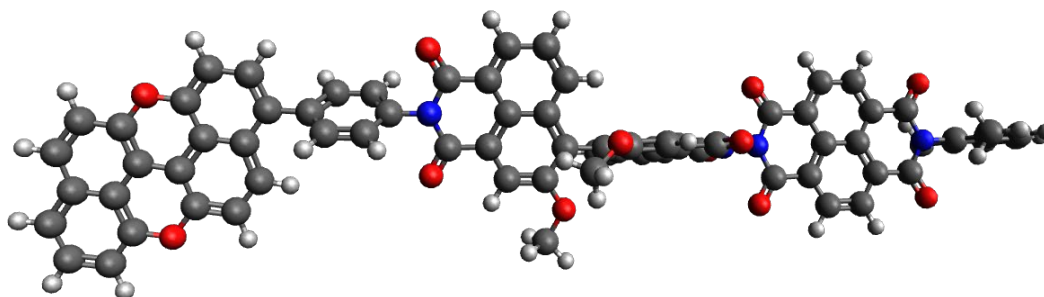
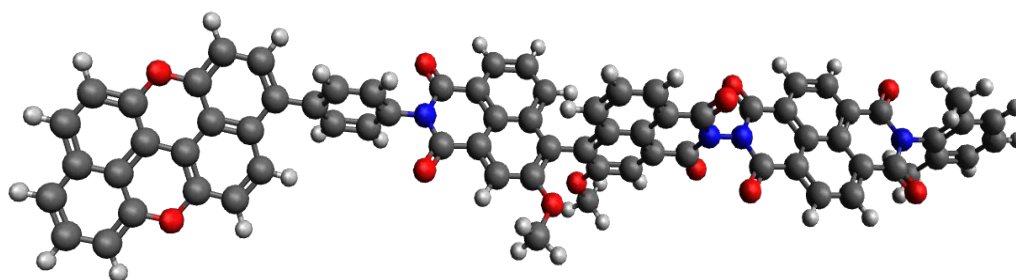


Fig. S8. OOP-ESEEM data and fits at $\theta = 0^\circ$ and $\theta = 90^\circ$ for the indicated molecules oriented in 5CB. Lower right: Hahn echo pulse sequence for OOP-ESEEM experiments, where T_{DAF} is the delay after the laser flash.

A



B



C

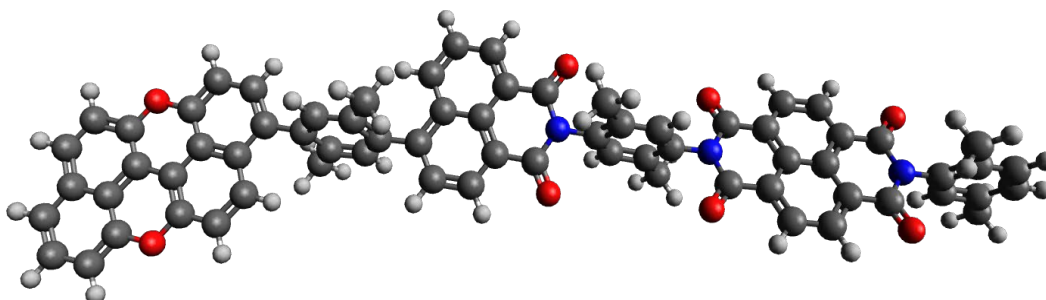


Fig. S9. Geometry optimized structures. (A) (*R*)-1-*h*₉, (B) (*S*)-1-*h*₉, and (C) 2-*h*₉ obtained using density functional theory (DFT) at the B3LYP/6-31G* level of theory in QChem (version 5.1).

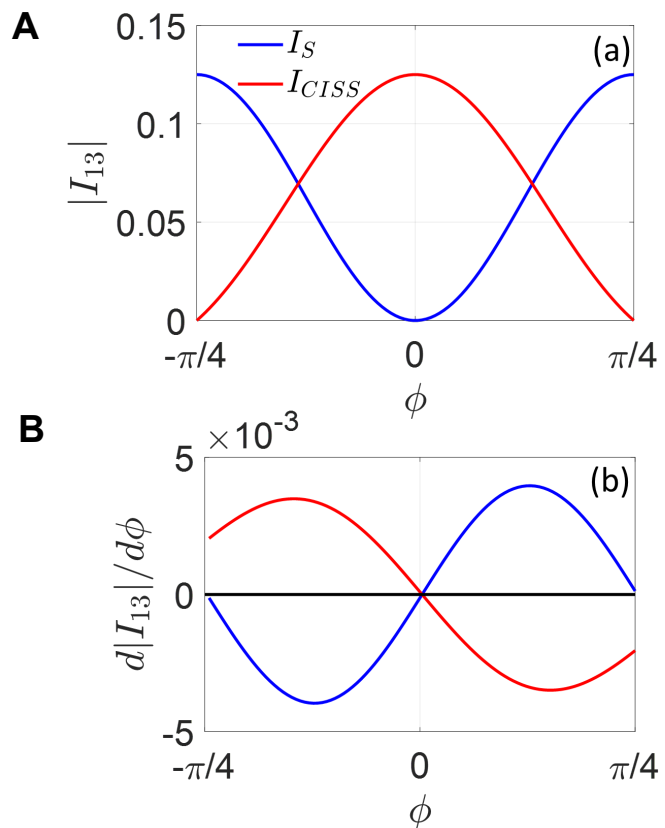


Fig. S10. Transition intensity as a function of the CISS contribution. (A) Dependence of the absolute value of the intensity of the $|\Phi_B\rangle \leftrightarrow |T_{+1}\rangle$ (1-3) transition on the composition of the eigenstates, expressed by the angle ϕ , for an initial singlet (blue line) or fully polarized (100% CISS, red curve) state. (B) Same for its derivative.

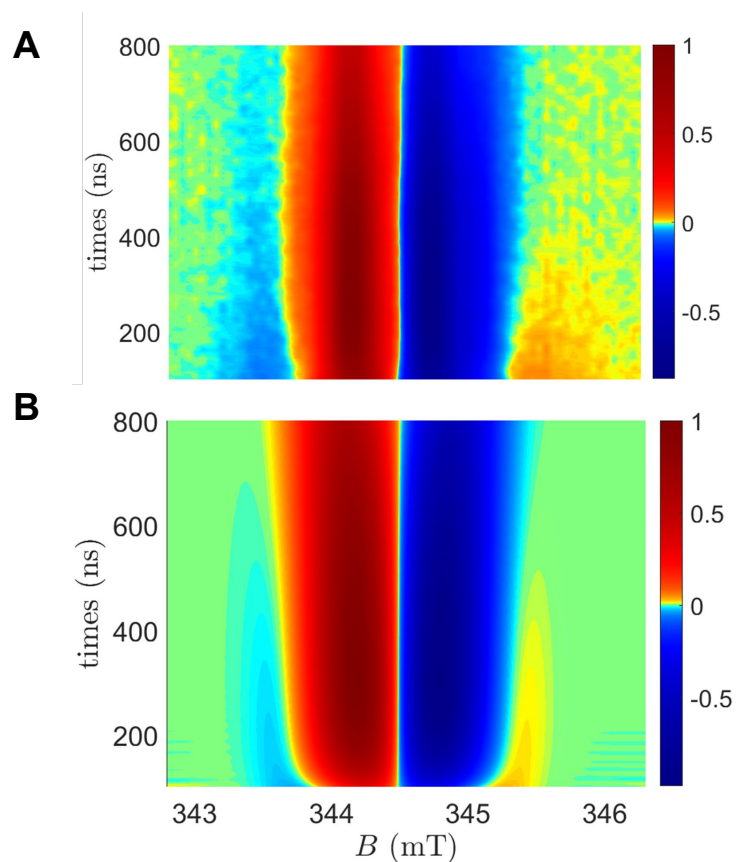


Fig. S11. Time and field dependence of TREPR spectra. Experimental (A) and simulated (B) time and field dependence of TREPR spectra of (*R* and *S*)-**1-h**₉ with the long axis of each molecule perpendicular to the applied magnetic field direction. The time dependence of the lateral CISS features is enhanced using a non-linear color-scale.

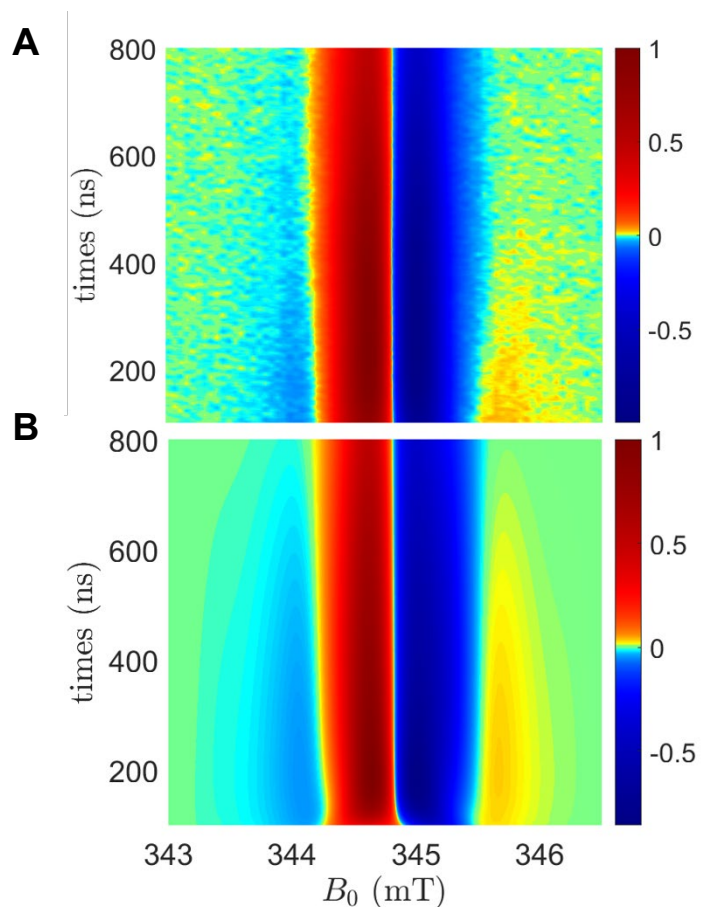


Figure S12. Time and field dependence of TREPR spectra. Experimental (A) and simulated (B) time and field dependence of TREPR spectra of (*R* and *S*)-1-*d*₉ with the long axis of each molecule perpendicular to the applied magnetic field direction. The time dependence of the lateral CISS features is enhanced using a non-linear color-scale. The effect of hyperfine interactions is included as a secular term in the Hamiltonian, with four ^1H ($I = 1/2$) and two ^{14}N ($I = 1$) nuclear spins coupled to $\text{NDI}^\cdot$ with isotropic $a_H = 4$ MHz and $a_N = 2$ MHz. For each configuration of the nuclear spins, we consider a distribution of hyperfine fields with a gaussian broadening of 1 MHz to account for possible strain and anisotropy of the hyperfine couplings. A distribution of donor-acceptor distances with standard deviation 0.5 nm was also included. Slightly different values of the g-tensors compared to the minimal model were used $g_{\text{NDI}} = (2.0034, 2.0041, 2.0043)$, $g_{\text{PXX}} = (2.0032, 2.0046, 2.0046)$ and a CISS contribution of 47 %.

Compound	Electron Transfer Reaction	r_{DA} (nm)	ΔG_{DA} (eV)	k_{ET} (s ⁻¹)	V_{DA} (cm ⁻¹)
(<i>R</i> or <i>S</i>) 1-<i>h</i>9 (<i>R</i> or <i>S</i>) 1-<i>d</i>9	¹ *PXX-NMI ₂ -NDI→ PXX ^{•+} -NMI ₂ ^{•-} -NDI	1.46 ^a	-0.26	8.1 x 10 ¹⁰ 8.3 x 10 ¹⁰	18 18
(<i>R</i> or <i>S</i>) 1-<i>h</i>9 (<i>R</i> or <i>S</i>) 1-<i>d</i>9	PXX ^{•+} -NMI ₂ ^{•-} -NDI→ PXX ^{•+} -NMI ₂ -NDI ^{•-}	2.48 ± 0.01 ^b	-1.09	4.9 x 10 ⁹ 1.5 x 10 ¹⁰	40 70
2-<i>h</i>9 2-<i>d</i>9	¹ *PXX-NMI-NDI→ PXX ^{•+} -NMI ^{•-} -NDI	1.21 ^a	-0.21	2.1 x 10 ¹¹ 2.6 x 10 ¹¹	74 83
2-<i>h</i>9 2-<i>d</i>9	PXX ^{•+} -NMI ^{•-} -NDI→ PXX ^{•+} -NMI-NDI ^{•-}	2.28 ± 0.01 ^b	-1.12	4.7 x 10 ⁹ 5.8 x 10 ⁹	18 20

Table S1. Electron Transfer Data. ^aDistances obtained from the DFT calculations given in Tables S3-S5. ^bDistances obtained from the OOP-ESEEM data given in the main text.

Compound	J (MHz)	r_{DA} (nm)
(<i>S</i>)- 1-h_9	-0.21 ± 0.09	2.48 ± 0.01
(<i>R</i>)- 1-h_9	-0.28 ± 0.11	2.48 ± 0.01
2-h_9	-0.14 ± 0.10	2.28 ± 0.01
(<i>S</i>)- 1-d_9	-0.31 ± 0.10	2.53 ± 0.01
(<i>R</i>)- 1-d_9	-0.18 ± 0.10	2.51 ± 0.01
2-d_9	-0.25 ± 0.10	2.29 ± 0.01

Table S2. OOP-ESEEM fit parameters

(R)-1h₉

Coordinates (Angstroms)

	ATOM	X	Y	Z
5	1 H	16.3358942944	-4.3281626064	-2.2877779347
	2 H	18.8019623042	-4.1693433187	-1.9479313717
	3 C	16.7611601820	-3.5145988373	-1.7095809383
	4 C	18.1587795359	-3.4119561690	-1.5083550495
	5 O	14.5660393913	-2.6739519154	-1.3610844712
10	6 H	11.9554419289	-2.7139744068	-1.5808736086
	7 C	15.9315444646	-2.5615966536	-1.1581652039
	8 H	6.7328227085	1.1956699080	2.5948679097
	9 C	12.3283682942	-1.8685093007	-1.0113134439
	10 C	18.7163833194	-2.3845926757	-0.7749051188
15	11 H	9.1962546309	1.1050068920	2.8402936355
	12 C	13.7165528883	-1.7324192465	-0.8150725568
	13 H	19.7921984943	-2.3275352957	-0.6334012584
	14 C	16.4771104203	-1.4935031639	-0.3987795457
	15 C	11.4606382462	-0.9370578779	-0.4711447059
20	16 H	-16.6065064634	-2.6405937900	2.4174190066
	17 C	7.3691531102	1.2034762054	1.7159872678
	18 C	-1.8269945346	-0.5152290510	0.2083806265
	19 C	17.8825534162	-1.3907657745	-0.1961869238
	20 C	-3.2551505378	-0.4921919533	0.1830938352
25	21 H	10.3942787638	-1.0680207988	-0.6182288983
	22 C	8.7541789852	1.1581587223	1.8496320274
	23 C	14.1997169495	-0.6752613331	-0.0733415615
	24 O	-6.0321160099	1.1785465171	-1.5033436950
	25 H	1.6450232955	-3.3923115486	-1.3680027496
30	26 C	-15.8988961538	-2.0526615079	1.8260684184
	27 C	15.6076544587	-0.5298791633	0.1585942148
	28 H	-15.2557354228	-1.5043821588	2.5250233216
	29 H	-15.2645627919	-2.7546285296	1.2714685306
	30 C	11.9189161302	0.1811046448	0.2893023948
35	31 C	18.3494238737	-0.2892156097	0.5804127404
	32 C	13.3273161708	0.2848801528	0.4886757962
	33 C	2.5580440635	-0.7477155995	-0.9855300313
	34 C	3.2308530614	0.3068324835	-0.3978768148
	35 C	4.7117015711	0.2278505312	-0.2925212371
40	36 C	3.2246847375	2.5106941690	0.6817524741
	37 C	2.5239245555	1.4266326195	0.0935108442
	38 C	4.7040872449	2.4832567201	0.7968925806
	39 C	1.1519314424	-0.7219852159	-1.0951769561
	40 C	2.5343610706	3.6077804483	1.1614267250
45	41 C	1.1250477687	-2.8658485725	-2.1775722624
	42 O	5.3427683922	-0.7345995100	-0.7007680978
	43 C	1.0964757572	1.4573806069	-0.0020593889
	44 C	1.1273931851	3.6459799453	1.0751780471
	45 C	0.4241377198	2.6007817471	0.5143995436

	46	C	0.4108253560	0.3559605914	-0.5997675805
	47	N	5.3586783224	1.3246848202	0.3099582008
	48	O	5.3426153314	3.3978093410	1.2893845788
	49	O	0.4381285903	-1.7149465830	-1.6950858853
5	50	C	6.7970869904	1.2652523619	0.4464860312
	51	C	-1.0813041063	0.3342850776	-0.6792930564
	52	H	3.1449148758	-1.5798425606	-1.3545876724
	53	C	-3.9199877312	0.3852762671	-0.7127643152
	54	C	-5.3988525559	0.4490882673	-0.7649716890
10	55	H	3.0942621527	4.4251178485	1.6031474860
	56	H	19.4179921686	-0.1870363654	0.7497114800
	57	C	16.0969213299	0.5186390063	0.9073452845
	58	C	9.5925612131	1.1646275382	0.7234413754
	59	H	0.3521351747	-3.5129393114	-2.5943785084
15	60	H	0.5916391273	4.5093864749	1.4583611731
	61	H	-18.5847437669	-1.7471402971	1.5416830154
	62	H	-0.6578046941	2.6441050410	0.4623305454
	63	C	11.0714018476	1.1706564492	0.8914466326
	64	C	17.4853523455	0.6445700690	1.1210554482
20	65	C	13.8817253115	1.3359251312	1.2666464753
	66	C	-16.6350630992	-1.1167235119	0.8979254592
	67	C	-18.0334175847	-1.0858628676	0.8780366665
	68	O	15.2466658104	1.4558545745	1.4589305555
	69	H	1.8445339355	-2.6047087247	-2.9631668082
25	70	C	11.6596541329	2.1660526720	1.6594099405
	71	C	7.6098222708	1.2742443630	-0.6864641475
	72	C	-1.7786537054	1.1604887390	-1.5592304434
	73	H	17.8546567885	1.4771307987	1.7115359085
	74	C	8.9942693199	1.2183667183	-0.5466360004
30	75	C	-3.1921607771	1.1886987654	-1.5616667162
	76	C	13.0536496921	2.2657897712	1.8551489924
	77	N	-6.0600321054	-0.4226177944	0.1289480289
	78	C	-8.0720450190	-1.2775784878	-0.7953395931
	79	C	-8.0587675521	0.5224447933	0.9917241579
35	80	O	-7.4190137550	-2.0155396053	-1.5056271643
	81	N	-7.4419081151	-0.3896705828	0.1055825028
	82	O	-7.3947108199	1.2341969395	1.7182648899
	83	C	-9.5554276085	-1.2266232375	-0.7924381444
	84	C	-9.5425210188	0.5283720875	0.9540374904
40	85	C	-10.2666136387	-2.0587509152	-1.6416549417
	86	C	-10.2396610820	-0.3368914872	0.0735730623
	87	C	-10.2413279431	1.3857385175	1.7882821826
	88	C	-11.6733486674	-2.0321755363	-1.6542399872
	89	C	-11.6615519420	-0.3124059291	0.0594812100
45	90	C	-11.6481530585	1.4075665373	1.7729162819
	91	C	-12.3661936008	-1.1733966689	-0.8166332991
	92	C	-12.3534243734	0.5717515708	0.9225113922
	93	C	-13.8547414752	-1.1637687658	-0.8420941530

	94	C	-13.8417432800	0.6077046860	0.9240729942
	95	H	-9.7168837727	-2.7294739324	-2.2935061538
	96	O	-14.4982786817	-1.8899765993	-1.5810889678
	97	H	-9.6819744803	2.0367884286	2.4518067997
5	98	N	-14.4962387750	-0.2696743019	0.0372911183
	99	O	-14.4746688691	1.3535231217	1.6525687876
	100	C	-15.9494855099	-0.2535365507	0.0320715017
	101	H	-12.2370578485	-2.6812570916	-2.3158569210
	102	H	11.0205994208	2.9199799813	2.1104896529
10	103	H	-12.2021268922	2.0753583226	2.4240462811
	104	C	-18.7183666280	-0.2241967210	0.0245121917
	105	H	13.4782569082	3.0670740983	2.4505982904
	106	H	7.1606246412	1.3273192063	-1.6726538926
	107	H	-19.8049880737	-0.2126463569	0.0215495427
15	108	H	9.6213876730	1.2450920080	-1.4330661484
	109	C	-5.4722302084	-1.3233034767	1.0492664600
	110	C	-16.6116549564	0.6239506080	-0.8372798088
	111	C	-18.0105375461	0.6227323822	-0.8250945128
	112	O	-6.1716778923	-2.0185844561	1.7609092303
20	113	C	-15.8511161853	1.5451770774	-1.7603172858
	114	H	-18.5437857318	1.2958553438	-1.4916735287
	115	H	-15.2088283665	2.2361038363	-1.2009031220
	116	H	-15.2123475188	0.9847053983	-2.4535991740
	117	H	-16.5432251711	2.1452427082	-2.3577702042
25	118	C	-3.9925083722	-1.3309211117	1.0605510605
	119	C	-3.3400425239	-2.1681109004	1.9475446532
	120	H	-3.9317792663	-2.7964165193	2.6048264370
	121	C	-1.9338814917	-2.1837520132	1.9893229627
	122	H	-1.4254090223	-2.8346269061	2.6945049820
30	123	C	-1.1967994922	-1.3781139013	1.1451306973
	124	H	-0.1140547656	-1.3958543095	1.1976229668
	125	O	-1.1924185285	2.0348260952	-2.4318562869
	126	C	-0.1820177022	1.5633367239	-3.3307740400
	127	H	-0.2906188841	2.1543016481	-4.2438657364
35	128	H	-0.3248378460	0.5022055350	-3.5609310127
	129	H	0.8199678125	1.7168091366	-2.9192293300
	130	H	-3.7019977033	1.8525417974	-2.2509995110

Table S3. Optimized cartesian coordinates for (R)-1-h₉.

(S)-1-*h*₉

Coordinates (Angstroms)

	ATOM	X	Y	Z
5	1 C	17.4475492751	-1.7348908307	0.4157596817
	2 C	18.3988117941	-0.7584104561	0.6446031257
	3 C	18.0515751565	0.6175297370	0.7872092214
	4 C	16.6717480335	0.9529257366	0.6810054414
	5 C	15.7123688121	-0.0564053915	0.4446238925
10	6 C	16.0871814926	-1.3764790487	0.3159447150
	7 C	18.9788188713	1.6668239536	1.0264080357
	8 C	18.5336977517	2.9670751178	1.1500370600
	9 C	17.1617779316	3.3006831816	1.0455022680
	10 C	16.2430600744	2.2993790511	0.8133484350
15	11 C	14.3346873559	0.3301272277	0.3455613577
	12 C	13.3723245058	-0.6791075337	0.1128919782
	13 C	13.8101192750	-2.0253690700	0.0031291594
	14 O	15.1490722085	-2.3638475260	0.0915262602
	15 O	14.9034385863	2.6385697819	0.7165072235
20	16 C	13.9660633291	1.6513072205	0.4870169573
	17 C	11.9893752169	-0.3446129856	0.0093429096
	18 C	11.0472652202	-1.4037220693	-0.2181172140
	19 C	11.5240049830	-2.7048787962	-0.3006010101
	20 C	12.8922482857	-3.0329406261	-0.1942869281
25	21 C	12.6049329114	2.0052299597	0.4114034295
	22 C	11.6504307121	1.0312654018	0.1823731182
	23 C	9.5863189140	-1.1584891203	-0.3634120439
	24 C	9.0756958417	-0.2777075887	-1.3322511746
	25 C	7.7023752498	-0.1067355114	-1.4936929360
30	26 C	6.8174365157	-0.8170310671	-0.6854696566
	27 C	7.3014025771	-1.6920909986	0.2844916947
	28 C	8.6743732683	-1.8584342995	0.4430025136
	29 N	5.3887179256	-0.6660113871	-0.8535014086
	30 C	4.7621586781	-1.4877111055	-1.8189458784
35	31 C	3.2854262795	-1.3812765205	-1.9191900076
	32 C	2.5678028776	-0.4900275506	-1.0801917529
	33 C	3.2515254092	0.3278566438	-0.1530256495
	34 C	4.7296037003	0.2562216743	-0.0178694306
	35 O	5.4216974499	-2.2452732765	-2.5109689966
40	36 O	5.3530645094	0.9412485843	0.7778335736
	37 C	1.1420384611	-0.4185973272	-1.1613057911
	38 C	0.4346007230	0.4675083554	-0.2985590738
	39 C	1.1468820722	1.2712749241	0.5947980634
	40 C	2.5549961935	1.1978191745	0.6648481162
45	41 C	2.6132762276	-2.1766744476	-2.8279087264
	42 C	1.2074975816	-2.1073226260	-2.9201577615
	43 C	0.4879956644	-1.2558979757	-2.1084739331
	44 C	-1.0577420273	0.5323103181	-0.3188832827
	45 C	-1.7021744722	1.4455479032	-1.1558118252

	46	C	-3.1110920305	1.5141474661	-1.1944879114
	47	C	-3.8747327062	0.6798541851	-0.3992732154
	48	C	-3.2619190434	-0.2547623298	0.4678266233
	49	C	-1.8343579021	-0.3293464538	0.5103923371
5	50	C	-4.0392107839	-1.1158772009	1.2890261010
	51	C	-3.4265053189	-2.0243837762	2.1315388757
	52	C	-2.0197108700	-2.1010120213	2.1798699893
	53	C	-1.2434490984	-1.2781852863	1.3916400881
	54	C	-5.3488575157	0.7838084590	-0.4860238454
10	55	N	-6.0554585536	-0.1030774820	0.3554174820
	56	C	-5.5166387885	-1.0590521727	1.2520664176
	57	O	-6.2546813072	-1.7596046036	1.9178411796
	58	O	-5.9433709589	1.5576339823	-1.2131537428
	59	O	-0.8976794367	2.2348889404	-1.9166128412
15	60	C	-1.4936369051	3.2053563147	-2.7701812683
	61	O	0.4042046792	2.0951401476	1.3827776473
	62	C	1.0685484472	2.9254224701	2.3287594685
	63	N	-7.4350343215	-0.0418599949	0.2880648814
	64	C	-8.0503772911	-0.8970684392	-0.6539480970
20	65	C	-9.5323645822	-0.8271426577	-0.6917617471
	66	C	-10.2295820994	0.0530770764	0.1735829007
	67	C	-9.5469350892	0.8893599445	1.0925203258
	68	C	-8.0652148510	0.8584508126	1.1767103480
	69	C	-10.2294286060	-1.6319106921	-1.5782343926
25	70	C	-11.6348491093	-1.5873563794	-1.6290957796
	71	C	-12.3403044195	-0.7372890410	-0.7931168807
	72	C	-11.6500757973	0.0967495386	0.1196816061
	73	C	-12.3550609353	0.9719459099	0.9812601774
	74	C	-11.6637604089	1.7810248578	1.8681869488
30	75	C	-10.2583679551	1.7392339855	1.9237758091
	76	C	-13.8274720935	-0.7105048362	-0.8575569077
	77	N	-14.4826830471	0.1715738678	0.0239695510
	78	C	-13.8425366912	1.0245799125	0.9442652295
	79	O	-7.4137781582	1.5423911054	1.9403863999
35	80	O	-7.3863304395	-1.6243780043	-1.3651353746
	81	O	-14.4863080466	1.7636507284	1.6701717773
	82	O	-14.4587532940	-1.4137301641	-1.6285887437
	83	H	13.2287803268	-4.0608672686	-0.2782598903
	84	H	10.8147273477	-3.5074012834	-0.4817839433
40	85	H	10.6061877074	1.3215717718	0.1476655403
	86	H	12.3200154972	3.0441649731	0.5444095140
	87	H	17.7277436849	-2.7783561345	0.3112943256
	88	H	19.4450214390	-1.0427783655	0.7201657738
	89	H	20.0369729431	1.4350585457	1.1112450688
45	90	H	19.2481993232	3.7646734310	1.3338029655
	91	H	16.8241839516	4.3267990303	1.1462052377
	92	H	9.7608276840	0.2566750838	-1.9837265800
	93	H	7.3153459614	0.5635634847	-2.2549578969

	94	H	6.6032415652	-2.2364365495	0.9131753330
	95	H	9.0510635008	-2.5316171771	1.2075068357
	96	H	3.1244851560	1.8043788300	1.3583326663
	97	H	3.1862713883	-2.8497640098	-3.4568973491
5	98	H	0.6877197155	-2.7358659067	-3.6374597085
	99	H	-0.5935485540	-1.2175981218	-2.1857811257
	100	H	-1.5453319798	-2.8169111242	2.8444997194
	101	H	-4.0472344406	-2.6691643958	2.7446700507
	102	H	-3.6299111919	2.2107281860	-1.8415862918
10	103	H	-9.6698752605	-2.2959764226	-2.2285579442
	104	H	-12.1876952227	-2.2159803368	-2.3190106865
	105	H	-12.2276333757	2.4423065219	2.5174649976
	106	H	-9.7100120229	2.3676272989	2.6176122581
	107	H	-0.6615671309	3.7180389509	-3.2550116658
15	108	H	-2.0871745086	3.9308496949	-2.2005969918
	109	H	-2.1258632051	2.7360948454	-3.5341004328
	110	H	0.2776681038	3.4769638426	2.8391938646
	111	H	1.7472158141	3.6325006437	1.8361662373
	112	H	1.6297952489	2.3324710646	3.0614662987
20	113	C	-15.9354808633	0.1958305600	-0.0122703144
	114	C	-16.5758042831	1.1033784824	-0.8669327383
	115	C	-15.7928075366	2.0518727685	-1.7422349671
	116	C	-17.9746602124	1.1066726916	-0.8850872666
	117	H	-18.4915017884	1.8025894620	-1.5411198600
25	118	C	-18.7030410777	0.2356850308	-0.0782102109
	119	H	-19.7893125171	0.2512253056	-0.1042471108
	120	C	-18.0395108006	-0.6552508471	0.7620667992
	121	H	-18.6075101702	-1.3352368184	1.3917472836
	122	C	-16.6420756964	-0.6920872765	0.8108531924
30	123	C	-15.9280367622	-1.6609388888	1.7224049060
	124	H	-15.1387130339	1.5127210430	-2.4381779245
	125	H	-15.1632655156	2.7231164956	-1.1456869805
	126	H	-16.4699504255	2.6725081560	-2.3358858926
	127	H	-15.2868386786	-1.1393352003	2.4429976006
35	128	H	-15.2954478879	-2.3551657409	1.1561325625
	129	H	-16.6490947469	-2.2564324861	2.2895347957
	130	H	-0.1618514242	-1.3475515026	1.4361573513

Table S4. Optimized cartesian coordinates for (S)-1-*h*9.

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2-h₉ Coordinates (Angstroms)

	ATOM	X	Y	Z
	1 C	16.2389988065	-1.2085745481	-0.2077704691
	2 C	17.1878902191	-0.2059808573	-0.2793452660
5	3 C	16.8299273794	1.1742023226	-0.3170548233
	4 C	15.4410854736	1.4855056127	-0.2778226179
	5 C	14.4840152293	0.4491779158	-0.2063419279
	6 C	14.8694027171	-0.8737099890	-0.1706059345
	7 C	17.7547747970	2.2498694870	-0.3903938501
10	8 C	17.2985198747	3.5518039724	-0.4212719286
	9 C	15.9175781122	3.8614500529	-0.3816585818
	10 C	15.0010803418	2.8342174542	-0.3105513955
	11 C	13.0970326691	0.8124965246	-0.1726387368
	12 C	12.1384555570	-0.2234624144	-0.1015353339
15	13 C	12.5875519895	-1.5696994087	-0.0576249332
	14 O	13.9335760099	-1.8865876429	-0.0994500212
	15 O	13.6524933722	3.1509131251	-0.2718134021
	16 C	12.7171148087	2.1380387642	-0.2036224147
	17 C	10.7480550103	0.0871085227	-0.0637994574
20	18 C	9.8083244508	-0.9918796366	0.0109132387
	19 C	10.2952627338	-2.2888995449	0.0684673186
	20 C	11.6738815602	-2.5962097871	0.0326865914
	21 C	11.3488373573	2.4716038626	-0.1555441801
	22 C	10.3974970273	1.4700278653	-0.0848704088
25	23 C	8.3317252197	-0.7595181719	0.0504412720
	24 C	7.6012106376	-0.3776154336	-1.0944429734
	25 C	6.2142493683	-0.2425632951	-0.9742137100
	26 C	5.5296733140	-0.4788391285	0.2250569668
	27 C	6.2584542969	-0.8476047203	1.3740350935
30	28 C	7.6454380811	-0.9796441927	1.2522427729
	29 C	4.0436434343	-0.3120409793	0.2473807510
	30 N	-6.6617074487	-0.0871117428	0.0317899129
	31 C	-7.1989954716	1.1549239433	0.4329140564
	32 C	-8.6824337269	1.2765791978	0.4028423851
35	33 C	-9.4792319116	0.1863298400	-0.0227609054
	34 C	-8.8909751873	-1.0360746132	-0.4270176042
	35 C	-7.4118089135	-1.1949965597	-0.4147551044
	36 C	-9.2815200505	2.4639669370	0.7924199155
	37 C	-10.6826414153	2.5973324631	0.7704126737
40	38 C	-11.4830388262	1.5438361414	0.3577295753
	39 C	-10.8943243035	0.3209156682	-0.0450258287
	40 C	-11.6904827031	-0.7703307791	-0.4691171980
	41 C	-11.0915596918	-1.9579032258	-0.8584565509
	42 C	-9.6905937706	-2.0903281668	-0.8389550103
45	43 C	-12.9625439039	1.7029380639	0.3417135964
	44 N	-13.7111299304	0.5942052784	-0.1030819200
	45 C	-13.1733751428	-0.6498257526	-0.4960251677
	46 O	-6.8637861581	-2.2227158779	-0.7751323587

	47	O	-6.4756912410	2.0703474716	0.7842692923
	48	O	-13.8970289769	-1.5688623336	-0.8389560481
	49	C	-15.1546589828	0.7392745158	-0.1546807916
	50	O	-13.5120837213	2.7327688242	0.6951322293
5	51	C	3.4947464923	0.9611792279	0.2088086728
	52	C	2.1019230158	1.1671133462	0.2190926194
	53	C	1.2359920549	0.0904167814	0.2636186679
	54	C	1.7530819681	-1.2292276265	0.2908756845
	55	C	3.1668059714	-1.4460190393	0.2771093727
10	56	C	0.8712316853	-2.3401449643	0.3248409869
	57	C	1.3699699198	-3.6304160318	0.3363252085
	58	C	2.7607386249	-3.8514261209	0.3063589431
	59	C	3.6384023920	-2.7862875641	0.2747457507
	60	C	-0.2275802471	0.3369229905	0.2716789316
15	61	N	-1.0566731497	-0.8064092500	0.2993107798
	62	C	-0.5994632762	-2.1396610863	0.3439366068
	63	O	-1.3907526830	-3.0683567764	0.3976357743
	64	O	-0.7106143978	1.4572431884	0.2538963635
	65	C	-2.4898471990	-0.5990471072	0.2617886624
20	66	C	-3.2513804196	-0.7151961922	1.4315379359
	67	C	-4.6305922052	-0.5316386760	1.3077274606
	68	C	-5.2197029881	-0.2407047259	0.0810921212
	69	C	-4.4574502508	-0.1041530699	-1.0853433086
	70	C	-3.0795013495	-0.2952408689	-0.9620516290
25	71	C	-15.9194129176	0.5629222038	1.0044927766
	72	C	-17.3051238910	0.7184582179	0.8756971520
	73	C	-17.8968008267	1.0286485001	-0.3460874637
	74	C	-17.1231004237	1.1976762310	-1.5010089538
	75	C	-15.7394902015	1.0490573728	-1.3807963998
30	76	C	5.5898402420	-1.0765189868	2.7111253811
	77	C	8.2689661013	-0.1523716750	-2.4318897687
	78	C	-15.2913606761	0.2318866574	2.3353903073
	79	C	-17.7635765474	1.5187317461	-2.8311611111
	80	C	-5.0808412467	0.2132739417	-2.4212502578
35	81	C	-2.6294808604	-1.0459030744	2.7643896152
	82	H	16.5283274190	-2.2543837020	-0.1794858149
	83	H	18.2411927950	-0.4720913366	-0.3079015850
	84	H	18.8198435370	2.0373044837	-0.4218104211
	85	H	18.0112582311	4.3699210677	-0.4771319913
40	86	H	15.5718029689	4.8895139374	-0.4063840345
	87	H	12.0188211331	-3.6240983751	0.0710325439
	88	H	9.5852615723	-3.1089357649	0.1317488983
	89	H	9.3475263123	1.7404292283	-0.0328548659
	90	H	11.0587329631	3.5175765206	-0.1685301742
45	91	H	8.2228109938	-1.2577071673	2.1308429273
	92	H	5.6374337608	0.0341247450	-1.8537879620
	93	H	4.1597060550	1.8194832807	0.1795617311
	94	H	1.6856988906	2.1689789140	0.1948086659

	95	H	3.1420873980	-4.8683904838	0.3037284137
	96	H	0.6681482981	-4.4574409110	0.3619468023
	97	H	-2.4475344558	-0.2074632570	-1.8408175044
	98	H	-5.2627547834	-0.6206601743	2.1864888851
5	99	H	-9.2133519982	-3.0156829381	-1.1435479711
	100	H	-11.7247431077	-2.7788973905	-1.1776567537
	101	H	-8.6483949250	3.2843249520	1.1134612214
	102	H	-11.1590379521	3.5233938711	1.0741983013
	103	H	-15.1008663835	1.1732882006	-2.2512005937
10	104	H	-18.9775269834	1.1395286334	-0.4033115290
	105	H	-17.9302390061	0.5919175915	1.7563413867
	106	H	4.7088835600	-2.9620638588	0.2414674872
	107	H	4.9571173014	-0.2278118206	2.9955067481
	108	H	6.3367414518	-1.2182586100	3.4979741818
15	109	H	4.9445162717	-1.9634368845	2.7024520535
	110	H	8.8909949801	-1.0087279264	-2.7162848522
	111	H	8.9269716201	0.7248979684	-2.4193180398
	112	H	7.5234146608	0.0018706665	-3.2179373251
	113	H	-5.7028355985	1.1146625473	-2.3728422750
20	114	H	-5.7136727762	-0.6124845956	-2.7675856034
	115	H	-4.3083483820	0.3824725330	-3.1768905609
	116	H	-2.2492885833	-2.0740594439	2.7707459835
	117	H	-3.3636218934	-0.9506510898	3.5697020181
	118	H	-1.7903299813	-0.3791517595	2.9937817489
25	119	H	-14.6693244671	-0.6692519090	2.2773927887
	120	H	-16.0620546533	0.0555260727	3.0915518177
	121	H	-14.6568904844	1.0529112394	2.6901940205
	122	H	-17.0101458972	1.7247154823	-3.5977846523
	123	H	-18.4179315116	2.3957306539	-2.7593981379
30	124	H	-18.3799122012	0.6840476547	-3.1886897819

Table S5. Optimized cartesian coordinates for 2-*h*₉.

θ	I_{CISS}	I_S	I_{13}
0	$\frac{1}{4} \cos^2 \phi$	$\frac{1}{2} \sin^2 \phi \cos^2 \phi$	$\frac{1}{4} \cos^2 \phi (1 - \cos 2\phi \cos 2\chi)$
90	$-\frac{1}{8} \cos 2\phi \cos^2 \phi$	$\frac{1}{2} \sin^2 \phi \cos^2 \phi$	$\frac{1}{4} \cos^2 \phi (\cos 2\chi - \cos 2\phi \cos^2 \chi)$

Table S6. Summary of the different intensities for transition $1 \leftrightarrow 3$ assuming 100% CISS, pure singlet and partial CISS efficiency (see Eqs. S16, S15, S13).

g_{NDI}	(2.0034, 2.0041, 2.0044)
g_{PXX}	(2.0031, 2.0044, 2.0046)
r_{DA}	2.5 nm
σ_D	0.35 nm
a_{NDI}	6.3 MHz
a_{PXX}	10 MHz
ΔB_0	0.4 mT

Table S7. Summary of the parameters used for the simulations of Fig. 4, where σ_D is the width of the gaussian distribution of DA distances (used to compute $D(\theta)$, Eq. S2) and ΔB_0 is the additional gaussian broadening used to convolute the spectra and accounting for inhomogeneous broadening of the remaining parameters. Parameters are given in the laboratory frame, with the magnetic field along z.

References and Notes

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